2026 at a Glance

Built 12 Nov 2026 from every BMT paper in the official archive (berkeley.mt/archives, 2012-2025): 1,058 short-answer problems tagged by topic, plus all 14 Power Rounds. Tags are one-topic judgment calls. The 2026 rows are forecasts from past patterns, not leaked or official content.

Round	Expected 2026 format	What the topics will most likely be	Confidence
Power (team, 90 min)	4 sections, about 40-45 questions, about 180-220 points. Mostly 'prove/show', with about 1 in 6 questions answer-only ('compute', 'find', 'draw'). Hands-on items possible (2024 used paper folding, 2025 pipe cleaners).	One topic new to almost everyone. My top guesses, by family: number theory / cryptography (unused since 2019), algebra and symmetry (unused since 2016), probability or games. Geometry-topology just ran twice (origami, knots), so less likely. The actual topic is announced only on the day.	Format: high. Topic: low
General (individual, 90 min, 25 Q)	25 problems; harder ones are often shared with the Focus tests. No calculus.	About 9 algebra (word problems are the most common single topic; then equations, floor, sequences), 6 geometry (areas, circles, triangles), 5 combinatorics (probability is the second most common topic), 5 number theory (digits/bases, mod). Problems 1-8 are warm-ups; 18-25 lean to circles, number theory and probability.	High
Algebra Focus (60 min, 10 Q)	10 problems, roughly easy to olympiad-flavoured	Polynomials & Vieta is the anchor (2-3 problems, usually #5-#10). Expect a sequence problem near #3, a floor/ceiling problem near #4, at least one roots-of-unity or complex problem in #7-#10.	High
Geometry Focus (60 min, 10 Q)	10 problems	Circles dominate (about 3-4 problems: tangency, power of a point, circumcircles). One 3D problem around #6-#8. #9-#10 are olympiad-style triangle-centre and circle configurations.	High
Discrete Focus (60 min, 10 Q)	10 problems, about 5-6 combinatorics / 4-5 number theory	Counting and probability in #1-#5; an expected-value or recursion problem near #7; arithmetic functions (divisors, sigma, gcd/lcm) near #8; a mod-prime sum, a graph or a Diophantine problem at #9-#10.	High
Calculus Focus (60 min, 10 Q)	10 problems	#1-#3 straightforward derivatives/integrals; limits and substitution in the middle; #8-#10 are 'trick' integrals (log-trig, symmetry, series expansion) or limits of sums. Derivatives ~25%, integration ~35%, limits ~15%.	High
Guts (team, 75 min, 27 Q)	9 sets x 3; points per question 10, 11, 12, 13, 14, 16, 18, 21, 25 (max 420)	Every set mixes subjects - usually one geometry, plus algebra/number theory/combinatorics. Set 9 is three estimation problems scored by closeness. No calculus beyond a rare one-off.	High
Tiebreakers (15 min)	5 questions (General) or 3 (each Focus); scored on correctness, then submission time	Same subject mix as the matching test, at its harder end.	High
Recurring motif	The year usually shows up	11-18 problems a year (counting General repeats) used the contest year. Know 2026 = 2 x 1013 (1013 is prime), 2026 = 45^2 + 1, sigma(2026) = 3042, phi(2026) = 1012.	Medium

Format History

From the headers of each archived paper. Problem counts are what was tagged; 2017's Geometry problem file is missing from the archive.

Year	Date	Individual-style tests	Subject tests	Team rounds	Power Round	Change worth knowing
2012	Spring 2012	Individual (10)	-	Team (10); Tournament round (7 rounds x 6)	Perfect matchings & recurrences	Original format
2013	23 Mar 2013	Individual (20)	Analysis, Discrete, Geometry (10 + 2 proof problems)	Team (10)	Noncommutative operators; 60 min, 100 pts	Subject tests appear, including calculus ('Analysis')
2014	22 Mar 2014	Individual (20)	Analysis, Geometry (10 + 2 proofs)	Team (20)	Symmetry groups of regular polyhedra; 60 min, 125 pts	
2015	7 Mar 2015	Individual (20)	Analysis, Geometry (10 + 2 proofs)	Team (20)	Projective planes; 60 min, 100 pts	Tiebreakers published from here
2016	12 Mar 2016	Individual (20)	Analysis, Discrete, Geometry (10)	Team (15)	Orthogonal polynomials; 60 min, 100 pts	Proof problems dropped from subject tests
2017	12 Mar 2017	Individual (20)	Analysis, Discrete, Geometry (10)	Team (15)	Tropical geometry; 60 min, 80 pts	
2018	18 Mar 2018	-	Algebra, Calculus, Combinatorics, Geometry, Number Theory (10)	Team (15)	Trees & tree encodings	Five focus subjects; no general test
2019	7 Apr 2019	Individual (20)	Algebra, Discrete, Geometry (10)	Team (15)	The arithmetic derivative; 60 min, 70 pts	Combinatorics + NT merged into 'Discrete'
2020	3-4 Oct 2020 (online)	Individual (25)	Algebra, Calculus, Discrete, Geometry (10)	Team (27, in sets)	Mathematical billiards; 1 week, 150 pts	Moved to autumn; today's four Focus tests fixed
2021	20-21 Nov 2021	General (25)	Algebra, Calculus, Discrete, Geometry (10)	Guts (9 x 3)	Combinatorial species & generating functions; 120 pts	Current format begins: General vs two Focus, Guts
2022	5 Nov 2022	General (25)	Same four (10)	Guts (9 x 3)	Error-correcting codes; 4 sections, 228 pts	Power Rounds get much longer
2023	4 Nov 2023	General (25)	Same four (10)	Guts (9 x 3)	Zero-knowledge proofs; 4 sections, 145 pts	Guts set 8 raised from 20 to 21 points
2024	2 Nov 2024	General (25)	Same four (10)	Guts (9 x 3)	Origami; 90 min, 4 sections, 185 pts	Power = 90 min; physical folding answers
2025	8 Nov 2025	General (25)	Same four (10)	Guts (9 x 3)	Knot theory; 90 min, 4 sections, 205 pts	Pipe-cleaner knots handed out
2026 (forecast)	14 Nov 2026	General (25)	Algebra, Calculus, Discrete, Geometry (10)	Guts (9 x 3)	Unknown topic; 90 min, ~4 sections, ~180-220 pts	Official page confirms the same three rounds and counts

Power Round History

Topics read from each archived Power Round. 'Family' is my grouping, used for the 2026 guess.

Year	Topic	Family	Structure / notes
2012	Perfect matchings and recurrences	Graphs & combinatorics	Graph definitions, counting matchings, recurrences
2013	The algebra of noncommutative operators	Algebra (abstract)	Motivated by quantum mechanics
2014	Symmetry groups of regular polyhedra	Algebra (abstract)	Group theory from scratch
2015	Projective planes	Geometry & topology	Finite geometries without parallel lines
2016	Orthogonal polynomials	Algebra / analysis	16 questions, ~1/4 conceptual
2017	Tropical geometry	Geometry & topology	Piecewise-linear algebraic geometry
2018	Trees and tree encodings	Graphs & combinatorics	Labelled trees, encodings (Pruefer-style)
2019	The arithmetic derivative	Number theory	Bounds and 'arithmetic differential equations'
2020	Mathematical billiards	Geometry & dynamics	Online, one week, outside tools allowed
2021	Combinatorial species and generating functions	Graphs & combinatorics	Trees, permutations, partitions, species
2022	Error-correcting codes	Computer science / information	Hamming, Singleton, Reed-Solomon bounds; 43 questions
2023	Zero-knowledge proofs	Computer science / information	Graph isomorphism, interactive proofs; 38 questions
2024	Origami	Geometry & topology	Constructions, flat folding (Kawasaki, Maekawa), solving equations by folding; 39 questions
2025	Knot theory	Geometry & topology	Reidemeister moves, invariants (tricolorability), rational knots, unknotting; 44 questions

Power 2026 Guess

My estimate only. Weighting: families unused for longest get more weight, the family used in both 2024 and 2025 gets less, and recent rounds favour visual/hands-on topics that start from zero background.

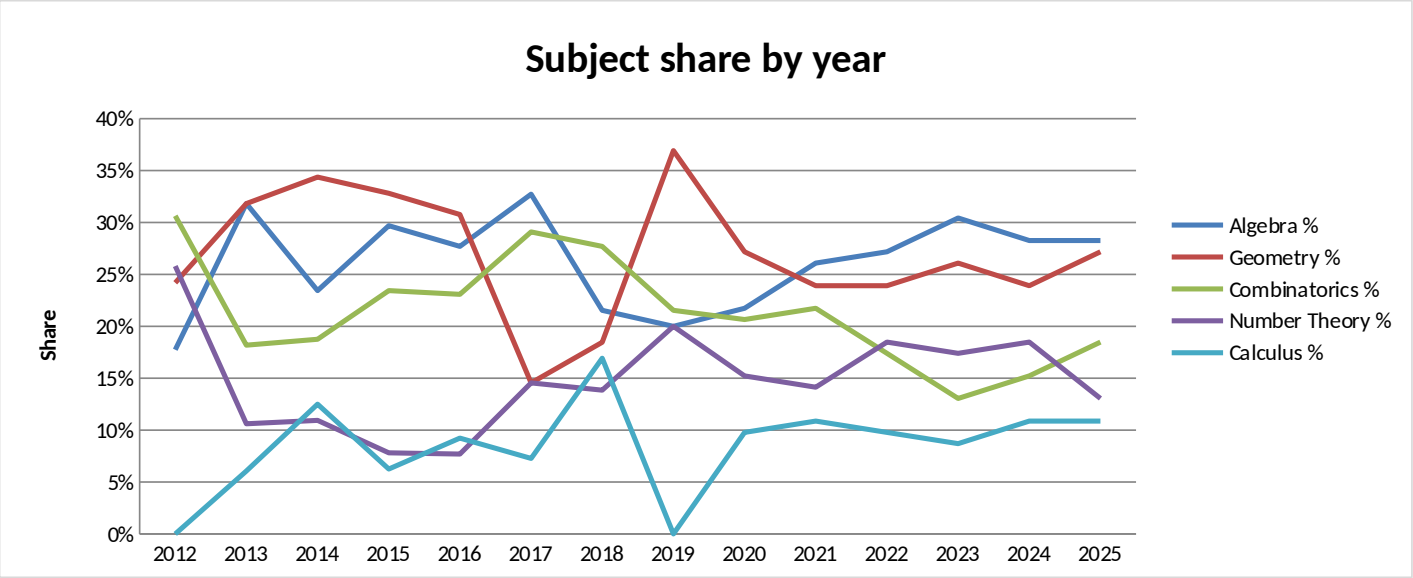
Topic family	Last used	Years since	My chance (%)	Example topics that would fit	What to review
Number theory / cryptography	2019	7	22	RSA and modular exponentiation, continued fractions, p-adic numbers, sums of squares, elliptic curves over finite fields	Mod arithmetic, Fermat/Euler, gcd and Bezout, induction
Algebra & symmetry	2016	10	18	Groups acting on objects, Burnside/Polya counting, permutation puzzles (Rubik-type), finite fields	Permutations, cycle notation, functions/bijections
Probability, games & dynamics	2020	6	18	Random walks, Markov chains, combinatorial games (Nim, Sprague-Grundy), voting/fair division, chip-firing	Expected value, recursion, invariants
Graphs & combinatorics	2021	5	16	Ramsey theory, graph colouring, matroids, tilings, posets, Young tableaux	Graph vocabulary, bijections, pigeonhole, induction
Computer science / information	2023	3	14	Sorting networks, complexity, automata, compression, voting algorithms	Algorithms as proofs, invariants, counting
Geometry & topology	2025	1	12	Tilings and polyhedra, convexity, Euler characteristic, billiards, curves of constant width	Euclidean basics, Euler's formula, symmetry
Total			**100**		

Whatever the topic, recent Power Rounds share a shape: Section 1 builds definitions with short 'compute/draw' warm-ups; Sections 2-3 prove the core theorems; Section 4 is a bigger result worth 15-25 points per question. Read everything first and split sections across the team.

Subject Trends

Count of tagged short-answer problems per year by subject (all tests incl. Guts/Team; Power excluded). Shares are formulas. 'Other' is mostly Guts estimation.

Year	Total	Algebra	Geometry	Combinatorics	Number Theory	Calculus	Other	Algebra %	Geometry %	Combinatorics %	Number Theory %	Calculus %	Other %
2012	62	11	15	19	16	0	1	18%	24%	31%	26%	0%	2%
2013	66	21	21	12	7	4	1	32%	32%	18%	11%	6%	2%
2014	64	15	22	12	7	8	0	23%	34%	19%	11%	13%	0%
2015	64	19	21	15	5	4	0	30%	33%	23%	8%	6%	0%
2016	65	18	20	15	5	6	1	28%	31%	23%	8%	9%	2%
2017	55	18	8	16	8	4	1	33%	15%	29%	15%	7%	2%
2018	65	14	12	18	9	11	1	22%	18%	28%	14%	17%	2%
2019	65	13	24	14	13	0	1	20%	37%	22%	20%	0%	2%
2020	92	20	25	19	14	9	5	22%	27%	21%	15%	10%	5%
2021	92	24	22	20	13	10	3	26%	24%	22%	14%	11%	3%
2022	92	25	22	16	17	9	3	27%	24%	17%	18%	10%	3%
2023	92	28	24	12	16	8	4	30%	26%	13%	17%	9%	4%
2024	92	26	22	14	17	10	3	28%	24%	15%	18%	11%	3%
2025	92	26	25	17	12	10	2	28%	27%	18%	13%	11%	2%
2021-25 avg								**28%**	**25%**	**17%**	**16%**	**10%**	**3%**



Topic Frequency

For each test: how often each topic appeared in 2021-2025 (the current format), and the expected number of 2026 problems = 2021-25 share x problems in the test. All-years share is shown for comparison. Filter column A to one test.

Test	Subject	Topic	2021-25 count	2021-25 problems	2021-25 share	2026 problems in test	Expected # in 2026	All-years count	All-years problems	All-years share
General	Algebra	Arithmetic & word problems	19	125	15%	25	3.8	25	280	9%
General	Combinatorics	Probability	12	125	10%	25	2.4	24	280	9%
General	Geometry	Polygons & area	11	125	9%	25	2.2	16	280	6%
General	Combinatorics	Counting & arrangements	10	125	8%	25	2.0	25	280	9%
General	Number Theory	Digits & bases	10	125	8%	25	2.0	14	280	5%
General	Algebra	Equations & systems	7	125	6%	25	1.4	10	280	4%
General	Geometry	Circles (tangents, power of a point, cyclic)	7	125	6%	25	1.4	21	280	8%
General	Geometry	Triangles (lengths, angles, areas)	6	125	5%	25	1.2	17	280	6%
General	Number Theory	Modular arithmetic	6	125	5%	25	1.2	13	280	5%
General	Algebra	Sequences, series & recurrences	6	125	5%	25	1.2	14	280	5%
General	Algebra	Floor, ceiling & absolute value	5	125	4%	25	1.0	6	280	2%
General	Number Theory	Divisibility, primes & factorization	5	125	4%	25	1.0	11	280	4%
General	Geometry	3D geometry	3	125	2%	25	0.6	11	280	4%
General	Number Theory	Arithmetic functions & valuations	3	125	2%	25	0.6	5	280	2%
General	Geometry	Similarity, ratios & mass points	2	125	2%	25	0.4	5	280	2%
General	Algebra	Polynomials & Vieta	2	125	2%	25	0.4	10	280	4%
General	Algebra	Complex numbers & roots of unity	2	125	2%	25	0.4	6	280	2%
General	Algebra	Logarithms & exponents	2	125	2%	25	0.4	4	280	1%
General	Algebra	Inequalities & optimization	2	125	2%	25	0.4	7	280	3%
General	Combinatorics	Inclusion-exclusion, bijections & lattice paths	1	125	1%	25	0.2	2	280	1%
General	Combinatorics	Expected value	1	125	1%	25	0.2	8	280	3%
General	Geometry	Transformations, loci & advanced tools	1	125	1%	25	0.2	4	280	1%
General	Number Theory	Diophantine equations	1	125	1%	25	0.2	4	280	1%
General	Combinatorics	Recursion & states	1	125	1%	25	0.2	1	280	0%
Algebra	Algebra	Polynomials & Vieta	11	50	22%	10	2.2	19	80	24%
Algebra	Algebra	Equations & systems	8	50	16%	10	1.6	10	80	13%
Algebra	Algebra	Sequences, series & recurrences	7	50	14%	10	1.4	12	80	15%

Test	Subject	Topic	2021-25 count	2021-25 problems	2021-25 share	2026 problems in test	Expected # in 2026	All-years count	All-years problems	All-years share
Algebra	Algebra	Floor, ceiling & absolute value	6	50	12%	10	1.2	6	80	8%
Algebra	Algebra	Functions & functional equations	4	50	8%	10	0.8	5	80	6%
Algebra	Algebra	Complex numbers & roots of unity	4	50	8%	10	0.8	6	80	8%
Algebra	Algebra	Logarithms & exponents	3	50	6%	10	0.6	3	80	4%
Algebra	Algebra	Trigonometry (algebraic)	2	50	4%	10	0.4	3	80	4%
Algebra	Algebra	Inequalities & optimization	2	50	4%	10	0.4	6	80	8%
Algebra	Algebra	Arithmetic & word problems	2	50	4%	10	0.4	3	80	4%
Algebra	Number Theory	Arithmetic functions & valuations	1	50	2%	10	0.2	1	80	1%
Geometry	Geometry	Circles (tangents, power of a point, cyclic)	17	50	34%	10	3.4	33	126	26%
Geometry	Geometry	Polygons & area	11	50	22%	10	2.2	21	126	17%
Geometry	Geometry	Triangles (lengths, angles, areas)	8	50	16%	10	1.6	23	126	18%
Geometry	Geometry	Transformations, loci & advanced tools	6	50	12%	10	1.2	14	126	11%
Geometry	Geometry	3D geometry	6	50	12%	10	1.2	20	126	16%
Geometry	Geometry	Similarity, ratios & mass points	1	50	2%	10	0.2	4	126	3%
Geometry	Geometry	Coordinate & analytic geometry	1	50	2%	10	0.2	8	126	6%
Discrete	Combinatorics	Counting & arrangements	11	50	22%	10	2.2	22	122	18%
Discrete	Combinatorics	Probability	8	50	16%	10	1.6	18	122	15%
Discrete	Number Theory	Modular arithmetic	6	50	12%	10	1.2	19	122	16%
Discrete	Number Theory	Digits & bases	6	50	12%	10	1.2	7	122	6%
Discrete	Number Theory	Arithmetic functions & valuations	6	50	12%	10	1.2	13	122	11%
Discrete	Combinatorics	Expected value	3	50	6%	10	0.6	8	122	7%
Discrete	Number Theory	Divisibility, primes & factorization	3	50	6%	10	0.6	6	122	5%
Discrete	Combinatorics	Recursion & states	2	50	4%	10	0.4	4	122	3%
Discrete	Combinatorics	Graphs, grids, colorings & tilings	2	50	4%	10	0.4	6	122	5%
Discrete	Number Theory	Diophantine equations	2	50	4%	10	0.4	5	122	4%
Discrete	Combinatorics	Inclusion-exclusion, bijections & lattice paths	1	50	2%	10	0.2	3	122	2%
Calculus	Calculus	Derivatives & optimization	13	50	26%	10	2.6	18	126	14%
Calculus	Calculus	Integration techniques	12	50	24%	10	2.4	15	126	12%
Calculus	Calculus	Limits	8	50	16%	10	1.6	17	126	13%
Calculus	Calculus	Advanced / tricky integrals	5	50	10%	10	1.0	11	126	9%
Calculus	Calculus	Applications (area, volume, rates)	4	50	8%	10	0.8	7	126	6%
Calculus	Calculus	Series (calculus)	4	50	8%	10	0.8	11	126	9%
Calculus	Algebra	Inequalities & optimization	3	50	6%	10	0.6	8	126	6%

Test	Subject	Topic	2021-25 count	2021-25 problems	2021-25 share	2026 problems in test	Expected # in 2026	All-years count	All-years problems	All-years share
Calculus	Combinatorics	Expected value	1	50	2%	10	0.2	1	126	1%
Guts	Other	Estimation	15	135	11%	27	3.0	18	324	6%
Guts	Geometry	Triangles (lengths, angles, areas)	10	135	7%	27	2.0	20	324	6%
Guts	Combinatorics	Probability	9	135	7%	27	1.8	21	324	6%
Guts	Number Theory	Digits & bases	9	135	7%	27	1.8	13	324	4%
Guts	Geometry	Circles (tangents, power of a point, cyclic)	8	135	6%	27	1.6	18	324	6%
Guts	Algebra	Equations & systems	7	135	5%	27	1.4	11	324	3%
Guts	Algebra	Arithmetic & word problems	6	135	4%	27	1.2	10	324	3%
Guts	Number Theory	Modular arithmetic	6	135	4%	27	1.2	13	324	4%
Guts	Combinatorics	Expected value	6	135	4%	27	1.2	24	324	7%
Guts	Geometry	Polygons & area	6	135	4%	27	1.2	14	324	4%
Guts	Number Theory	Divisibility, primes & factorization	5	135	4%	27	1.0	15	324	5%
Guts	Combinatorics	Counting & arrangements	5	135	4%	27	1.0	19	324	6%
Guts	Geometry	3D geometry	4	135	3%	27	0.8	10	324	3%
Guts	Geometry	Coordinate & analytic geometry	4	135	3%	27	0.8	10	324	3%
Guts	Algebra	Polynomials & Vieta	4	135	3%	27	0.8	10	324	3%
Guts	Number Theory	Arithmetic functions & valuations	4	135	3%	27	0.8	9	324	3%
Guts	Algebra	Sequences, series & recurrences	4	135	3%	27	0.8	16	324	5%
Guts	Algebra	Logarithms & exponents	3	135	2%	27	0.6	6	324	2%
Guts	Combinatorics	Graphs, grids, colorings & tilings	3	135	2%	27	0.6	8	324	2%
Guts	Algebra	Complex numbers & roots of unity	3	135	2%	27	0.6	4	324	1%
Guts	Geometry	Similarity, ratios & mass points	2	135	1%	27	0.4	5	324	2%
Guts	Combinatorics	Inclusion-exclusion, bijections & lattice paths	2	135	1%	27	0.4	5	324	2%
Guts	Number Theory	Diophantine equations	2	135	1%	27	0.4	7	324	2%
Guts	Algebra	Floor, ceiling & absolute value	2	135	1%	27	0.4	2	324	1%
Guts	Algebra	Trigonometry (algebraic)	1	135	1%	27	0.2	2	324	1%
Guts	Geometry	Transformations, loci & advanced tools	1	135	1%	27	0.2	5	324	2%
Guts	Algebra	Inequalities & optimization	1	135	1%	27	0.2	5	324	2%
Guts	Combinatorics	Games, invariants & pigeonhole	1	135	1%	27	0.2	7	324	2%
Guts	Algebra	Functions & functional equations	1	135	1%	27	0.2	5	324	2%
Guts	Calculus	Limits	1	135	1%	27	0.2	2	324	1%

Algebra Forecast

Slot-by-slot forecast for the 2026 Algebra Focus test (10 problems, 60 min). Built from what appeared in each slot 2021-2025 (shown on the right). Difficulty band is my reading of slot position and topic.

#	Difficulty	Most likely topic for 2026	Tools to have ready	What was in this slot 2021-2025
1	Easy	Linear equations / systems or a word problem	Set up and solve two equations	'21: Two quadratic equations in x, compute x^4; '22: Iterated linear operation; '23: Two lines meeting at a point, find slopes; '24: Apples word problem, linear equations; '25: Linear combination of two equations
2	Easy	Exponent/log equation, absolute value, or simple function	Substitute u = 2^x, case on signs	'21: Linear functions commutator difference; '22: Exponential equation as quadratic in 2^x; '23: Cases on |x|,|y| signs, |x-y|; '24: Rational equations with equal values, substitution; '25: Average with added 20 slips
3	Easy	Arithmetic and geometric sequences (4 of last 5 years)	a + (n-1)d, ar^(n-1), sum formulas	'21: Nested floor equation, upper bound of solutions; '22: Geometric then arithmetic sequence, minimize d/a; '23: Geometric sequences with equal ratio; '24: Arithmetic plus geometric sequences; '25: Arithmetic sequence ratio and sum of terms
4	Easy-Med	Floor / ceiling equation (3 of last 5 years)	Write x = n + f with 0 <= f < 1	'21: Sin 2θ equation, maximize (1+sin)(1+cos); '22: Floor with ceiling nested equation, intervals; '23: Piecewise linear f via floor, evaluate polynomial; '24: Floor and ceiling equation, greatest x; '25: Translate and rotate graph, roots of g
5	Medium	Vieta / symmetric sums of roots	e1, e2, e3; shifted roots	'21: Floor times fractional part equation; '22: Extremes of T given sum and sum of squares; '23: Vieta on shifted roots; '24: Cubic roots equidistant from 2024, unique pair; '25: Cube sum identities, symmetric system complex
6	Medium	Functional equation, polynomial values, or symmetric system	Plug in special values; finite differences	'21: Functional equation with transformations of x; '22: Degree-6 polynomial from difference values; '23: Recurrence generating function sum equals 1; '24: Symmetric three-variable quadratic system; '25: Sine of log expression, sum of solutions
7	Med-Hard	Vieta on a cubic, or roots of unity	sum 1/(1-r), product of (1 - w^k)	'21: Roots of unity sum of 1/(1-z^2020); '22: Vieta on cubic, sum of 1/(1-r^2); '23: Count polynomial pairs a^2+b^2 factorization; '24: Triples with symmetric equation, AM-GM equality; '25: Product of squared differences of fifth roots of unity
8	Hard	Polynomial / inequality optimisation	AM-GM, Cauchy-Schwarz, Lagrange-style reasoning	'21: Cubic roots as right triangle sides; '22: Products of shifted variables as polynomial values; '23: Cubic roots as right triangle sides, minimize t; '24: Min of linear form under symmetric constraints; '25: Change of base logs, simplify ratio
9	Hard	Trig recursion, complex numbers, or iterated maps	tan addition, cis, conjugate pairs	'21: Reducibility of x^n+x+1; '22: Tangent addition recurrence, first undefined term; '23: Iterating logistic map, count preimages; '24: Trig-substitution recursions, bounded differences; '25: Complex numbers with fifth power identity, abc values
10	Very hard	Products over roots of unity, power sums, polynomial factorisation	Newton sums, cyclotomic polynomials	'21: Square-free counting sums, sign of difference; '22: Vieta cyclic expressions on cubic; '23: Cubic form factors, integer solution triple; '24: Product over fifth roots of unity with sec, mod 101; '25: Power sum condition on ten complex numbers

Geometry Forecast

Slot-by-slot forecast for the 2026 Geometry Focus test (10 problems, 60 min). Built from what appeared in each slot 2021-2025 (shown on the right). Difficulty band is my reading of slot position and topic.

#	Difficulty	Most likely topic for 2026	Tools to have ready	What was in this slot 2021-2025
1	Easy	Areas of rectangles, semicircles, simple triangles	Pythagoras, area formulas	'21: Rectangle cut maximizing perimeter; '22: Paper folding, longest distance between corners; '23: Semicircle in rectangle, diagonal; '24: Three semicircles around equilateral triangle perimeter; '25: Right triangle with square, perimeter of pentagon
2	Easy	Circle through points on a square/polygon; area composite	Perpendicular bisectors; coordinates	'21: Smallest triangle containing six unit circles; '22: Tilted square circumscribing a square; '23: Circle through six trisection points; '24: Circle through trisection points of square; '25: Sheep tethered at corner, grazing area
3	Easy-Med	Composite areas, polygons, or a solid's volume	Regular polygon angles; frustum/cylinder	'21: Reflection for equal angles, AE+EB; '22: Tangent circles in circumcircle, shaded area; '23: Frustum plus cylinder volume; '24: Visibility region blocked by circle, tangent lines; '25: Square, pentagon, hexagon on common side, angle
4	Medium	Tangent circles, tangent lengths, ratios	Equal tangents; centre-to-centre distances	'21: Equilateral polygon with 15 and 300 degree angles; '22: Area swept by moving triangle in hexagon; '23: Tangent lengths, perimeter of triangle around circle; '24: Internally tangent circles, ratio with point on circle; '25: Equilateral triangle, perpendicular cevians, ratio
5	Medium	Circles inscribed in triangles/kites; areas	Inradius r = A/s, power of a point	'21: Two circles, common tangent, power of point; '22: Tilted tricycle wheels tracing circles; '23: Two tangent circles in right triangle; '24: Kite with circumcircle and incircle, area ratio; '25: Quadrilateral with distances to lines, area
6	Medium	Triangle centres, loci, similar triangles	Incenter/orthocenter facts; ellipse locus	'21: Shortest ant path on modified cube; '22: Incenter on line parallel to side, find AY; '23: Median extended to circumcircle, sine of central angle; '24: Orthic configuration areas, Ceva-like ratios; '25: Locus of centers of tangent circles, ellipse area
7	Med-Hard	3D geometry (tetrahedra, spheres, prisms)	Cross-sections, midsphere, coordinates in 3D	'21: Regular hexagon area from distances to line; '22: Orthocenter, angle bisector, find BC; '23: Tetrahedron midsphere radius; '24: Parallelogram altitudes, find CJ; '25: Prism lattice points, Pick-type volume
8	Hard	Triangle with orthocenter/circumcircle, or a 3D chord problem	Law of sines/cosines, extended law of sines	'21: Circumcircles through O, line perpendicular to BC; '22: Nested polygons and circles, limiting area; '23: Circle meets equilateral triangle, area difference; '24: Perpendicular chords in sphere, tetrahedron volume; '25: Orthocenter maximize angle, find AB+AC
9	Hard	Circles + power of a point, rotations, radical axes	Spiral similarity, radical axis	'21: Equilateral triangles, rotation, area of midpoint triangle; '22: Radical planes of seven spheres; '23: Isogonal-like point T, midpoint, find BC; '24: Incenter circle BIC, power of point from midpoint; '25: Circumcircles of midpoint triangles, radical axis, AK squared
10	Very hard	Olympiad configuration (orthic triangle, polars, circumcircles)	Inversion, poles/polars, trig bash	'21: Minimize sum of distances via reflections, incircle; '22: Tangent at A, orthic triangle line, find XY; '23: Circumcenter, circle BOC, perpendiculars, AF*EG; '24: Incircle chord EF, poles, polar tools, area; '25: Angle bisectors meeting circumcircles, XY parallel BC

Discrete Forecast

Slot-by-slot forecast for the 2026 Discrete Focus test (10 problems, 60 min). Built from what appeared in each slot 2021-2025 (shown on the right). Difficulty band is my reading of slot position and topic.

#	Difficulty	Most likely topic for 2026	Tools to have ready	What was in this slot 2021-2025
1	Easy	Basic counting, a probability, or a digit puzzle	Complementary counting	'21: Three-card hands with both colors; '22: Lunch choices with constraints; '23: Three-digit number divisibility and remainder; '24: Third-largest palindromic multiple of three; '25: Two socks same color
2	Easy	Divisors / counting with a constraint	Prime factorisation, tau(n)	'21: Two arithmetic progressions common terms; '22: Divisors of 100000 without digit 0; '23: Enter/exit orderings of names; '24: Periodic coloring with color usage limits; '25: n+10 multiple of n, factor of n+100
3	Easy-Med	Counting distributions or divisibility	Stars and bars, casework	'21: Distinct sums from 8 chosen numbers; '22: Dice sum versus coin product; '23: Count n where digit 4s increase on adding one; '24: Divisors of 44 with 24-n; '25: Distribute identical blocks, positive, repeated count
4	Medium	Probability, circular arrangements, colourings	Burnside-light, complementary counting	'21: Two players both captains; '22: Arm wrestling outcomes with more than 3 streaks; '23: Nested remainders mod 11,5,2 count; '24: Circular two-coloring with no isolated player; '25: Two-digit substrings divisible by three
5	Medium	Digits/palindromes, probability, gcd-lcm	Digit sums, mod 9/11	'21: Palindromic sum of number and reversal; '22: Digit sum equation for n; '23: Geometric N then N rolls, probability of 6; '24: Loaded dice with equal sum distribution; '25: gcd of lcms equals 2025, minimize sum
6	Medium	Divisor counts, bases, structured counting	tau(n^n), base-b expansions	'21: Number of divisors of n^n; '22: Quadratic with distinct real roots from powers of 2; '23: gcd lcm condition counting via CRT; '24: Base 6 permutation numbers divisible by 43; '25: Alternating multiples-of-three increasing sequences
7	Med-Hard	Expected value or a recursion/states problem	Linearity of expectation, set up E_k	'21: Minimum operations to reach 1; '22: Expected cards flipped until ace and king; '23: Expected extra area, random chord of square; '24: Probability of GIGI substring in arrangement; '25: Seat permutations to adjacent seats on circle
8	Hard	Arithmetic functions: sigma, tau, gcd/lcm (3 of last 5 years)	Multiplicativity, sigma parity	'21: Expected value of random nondecreasing decimal; '22: Linear recurrence pair mod 31; '23: Divisor-sum function iterates mod 1001; '24: Maximize ordered factorizations ratio over n; '25: Sum of divisors with even sigma, count h odd
9	Hard	Sums mod a prime, permutation counting, or graph extremal	Fermat, inverses mod p, Wilson	'21: Harmonic sum mod prime 101; '22: Selecting even-sized subsets, counting mod 42^2-1; '23: Counting permutations with three monotone sections; '24: Sum of squares mod prime 83; '25: Max edges with distinct degrees nonadjacent
10	Very hard	Graph theory, Diophantine, or grid arrangements	Turan, factoring, mod obstructions	'21: Triangle-free graph with 22 edges, Turan; '22: Pairs with a^2+b^2 dividing 10!; '23: Power of quadratic irrational mod prime; '24: 3x3 grid arrangements with adjacent sum repeats; '25: Perfect square 3^p*21^q+9*2^r, sum over triples

Calculus Forecast

Slot-by-slot forecast for the 2026 Calculus Focus test (10 problems, 60 min). Built from what appeared in each slot 2021-2025 (shown on the right). Difficulty band is my reading of slot position and topic.

#	Difficulty	Most likely topic for 2026	Tools to have ready	What was in this slot 2021-2025
1	Easy	A basic derivative (trig, product, FTC)	d/dx rules	'21: Fundamental theorem of calculus derivative; '22: Derivative of sin^2 plus cos^2; '23: Derivative of trig at pi/4; '24: Limit of cubic difference quotient; '25: Intersect f and f prime, solve quadratic
2	Easy	Higher derivatives or derivative conditions on a polynomial	Product rule, f'(a) = f'(b)	'21: Fourth derivative of product of linears; '22: Equal derivative values of cubic; '23: Integrate polynomial in (x-2), odd symmetry; '24: Derivative of inverse-like composed function; '25: Maximize x^pi style function over positives
3	Easy	Simple integral/area or derivative of a rational function	u-substitution, power rule	'21: Quadratic and derivative intersect once; '22: Substitution with nested exponentials; '23: Area under sqrt(kx); '24: Third derivative of rational function; '25: Limit of derivative at infinity
4	Medium	A limit, tangency, related rates or area between curves	Tangency: f = g and f' = g'	'21: Area of region bounded by parabolas; '22: Polynomial with shared roots of f and f'; '23: Related rates, melting cube; '24: Tangency of exponential and root, unique solution; '25: Limit of (2 arctan n/pi)^n
5	Medium	Integral with floor function or telescoping series	Split at integers	'21: Trig integral antiderivative matching; '22: Telescoping infinite sum of radicals; '23: Riemann sum lower bound on cubic form; '24: Integral of x^floor(x) asymptotics; '25: Integral with floor alternating signs
6	Medium	Limit or substitution integral	L'Hopital, standard limits	'21: Iterated sin(arccos) limit; '22: Substitution in trig integral; '23: nth derivatives of 1/x, sum series; '24: Integral with nested radicals, rational result; '25: Expected binomial count limit, geometric rolls
7	Med-Hard	Limit of a probability/area, integral by substitution, or series	Riemann sums, generating functions	'21: Integral of inverse function; '22: Limit of 1+integral to power 1/x; '23: Series sum n a_n/6^n via recurrence; '24: Integral by substitution with half powers; '25: Limit of circle area fraction above parabola
8	Hard	Tricky integral, alternating series, or optimisation over functions	Symmetry x -> a - x, logs from series	'21: Integral of t^(6n-1)/(1+t^3), limit; '22: Expected number of uniform purchases exceeding 1; '23: Maximize integral pointwise over f, completing square; '24: Alternating series telescoping to logs; '25: Sign of integral with sin and gamma, count k
9	Hard	Trig-power or log-trig integral, Newton's method, Riemann bounds	Reduction formulas	'21: Supremum of accumulated score, integral-like; '22: Integral of cot(x)ln(cos x); '23: Integral of (sin+cos)^6 over period; '24: Newton iteration basin convergence; '25: Riemann sum upper bound of cubic form
10	Very hard	Hard definite integral or infinite product/limit	ln(sin x) tricks, Feynman's trick, series of sin	'21: Infinite product with arctan; '22: Volume swept by rotating cube; '23: Integral of sin x/x^2 with series of sin(nx)/n!; '24: Recursive f with ln 2, average error limit; '25: Integral of ln(sin^4+cos^4)

General Forecast

The 2026 General Test (25 problems, 90 min). See 'Topic Frequency' (filter Test = General) for expected counts per topic. No calculus has appeared on the General test.

Problems	Difficulty	Most likely content	Subject mix in 2021-25
1-4	Easy	Word problems, arithmetic, a simple area or triangle	Algebra ~50%, Geometry ~30%
5-8	Easy-Med	Digits/bases, basic counting, first probability problem	Mixed; number theory appears
9-13	Medium	Equations, floor functions, sequences, probability	Algebra and combinatorics lead
14-17	Medium	Polygons/area, counting, divisibility, expected value	Even mix
18-21	Med-Hard	Circles, digits/bases, modular arithmetic, probability	Geometry and number theory lead
22-25	Hard	Probability, Diophantine, 3D, sequences - often a shared Focus problem	Spread across all four

Guts Forecast

9 sets x 3 problems, 75 minutes. Points per question have been identical since 2023 (set 8 was 20 in 2021-22). Max = 3 x sum of set points. Subject counts are 2021-25 totals (15 problems per set).

Set	Points / Q	Set total	Difficulty	Likely content	Subjects 2021-25	2025 examples
1	10	30	Easy	Quick warm-ups: word problem, factor/digit question, simple area	Algebra 6, Number Theory 4, Geometry 3, Combinatorics 2	'25: Perimeter of union of three squares | '25: Burger rate word problem system | '25: Arrange digits to minimize product expression
2	11	33	Easy	Easy mixed: factors, digit sums, a square or triangle	Geometry 5, Number Theory 5, Algebra 3, Combinatorics 2	'25: Six distinct factors of 30 with average 6 | '25: Digit sum operations then squared equals 2025 | '25: Multiples of 15 with digit sum above 15
3	12	36	Easy-Med	Colourings/counting, right-triangle geometry, sequences	Geometry 6, Combinatorics 4, Number Theory 3, Algebra 2	'25: Coloring six squares with adjacency constraints | '25: Right triangle with rectangle, area ratio, find CD | '25: Arithmetic and geometric sequences, equal terms
4	13	39	Medium	Dice/probability, an equation, a 3D or pyramid problem	Geometry 5, Combinatorics 4, Number Theory 4, Algebra 2	'25: Sum of two dice equal to winning numbers | '25: Cubic with floor equation, sum of cubes of solutions | '25: Pyramid with lateral edge angle, base area
5	14	42	Medium	Operations/puzzles, exponents, area ratios (cevians)	Algebra 6, Number Theory 4, Combinatorics 3, Geometry 2	'25: Nonassociative operation, max minus min over bracketings | '25: Exponent equals one, cases of base | '25: Cevian areas ratios, area of triangle PQR
6	16	48	Med-Hard	Mixed medium; rare calculus or base-b problem	Geometry 4, Algebra 4, Combinatorics 3, Number Theory 3, Calculus 1	'25: Digitwise units product operation, count a | '25: Rectangle with maximized quadrilaterals, area | '25: Palindromes in base 2025, smallest sqrt constant
7	18	54	Hard	Expected values, region counting, graph/gcd problems	Algebra 5, Geometry 5, Combinatorics 4, Number Theory 1	'25: Expected area of quadrilateral from random points | '25: Lines and parabola, count regions | '25: Connected graph by gcd, minimal trips
8	21	63	Hard	Hardest exact-answer set: Markov chains, products, solids	Geometry 4, Combinatorics 4, Algebra 4, Number Theory 2, Other 1	'25: Expected flips with reset Markov process | '25: Product of 1+1/a_j equals 2025 | '25: Volume of solid from net
9	25	75	Estimation	Estimation: three problems, partial credit by closeness	Other 14, Geometry 1	'25: Cyclic pentagon, inner pentagon area | '25: Count reducible short polynomials | '25: Sum of Euclidean algorithm step counts
Max		420				
Pacing		8.3		minutes per set if you aim to reach set 9		

Problem Log

Every tagged problem, 2012-2025. 'Test' maps old names onto today's (Individual -> General, Team/Tournament -> Guts, Analysis -> Calculus, Combinatorics/Number Theory -> Discrete). Use the filters.

Year	Test	Source file	#	Subject	Topic	Key idea
2012	General	IS2012	1	Algebra	Sequences, series & recurrences	Arithmetic progression, first negative term
2012	General	IS2012	2	Number Theory	Divisibility, primes & factorization	Smallest number with 28 divisors
2012	General	IS2012	3	Algebra	Sequences, series & recurrences	Telescoping infinite sum, largest prime factor
2012	General	IS2012	4	Combinatorics	Probability	Conditional probability with two dice
2012	General	IS2012	5	Number Theory	Modular arithmetic	Power tower mod 60
2012	General	IS2012	6	Geometry	Circles (tangents, power of a point, cyclic)	Incircle tangency points on diagonal, cyclic quadrilateral
2012	General	IS2012	7	Number Theory	Divisibility, primes & factorization	Distinct primes with constraints, maximize difference
2012	General	IS2012	8	Combinatorics	Expected value	Expected tosses to reach 60 heads in a row
2012	General	IS2012	9	Combinatorics	Games, invariants & pigeonhole	Bowling pins game, winning strategy
2012	General	IS2012	10	Geometry	Transformations, loci & advanced tools	Laser reflections in triangle, unfolding
2012	Guts	TS2012	1	Number Theory	Digits & bases	Sum of repeating base-6 expansions
2012	Guts	TS2012	2	Algebra	Logarithms & exponents	Telescoping product of logs
2012	Guts	TS2012	3	Geometry	Triangles (lengths, angles, areas)	Squares on triangle sides, area ratio
2012	Guts	TS2012	4	Combinatorics	Probability	Bayesian probability spy among missions
2012	Guts	TS2012	5	Number Theory	Divisibility, primes & factorization	Divisibility multiplier for 2013
2012	Guts	TS2012	6	Geometry	Circles (tangents, power of a point, cyclic)	Circles on diameters, area ratio
2012	Guts	TS2012	7	Algebra	Sequences, series & recurrences	Geometric series for meeting distance
2012	Guts	TS2012	8	Number Theory	Arithmetic functions & valuations	Iterated Euler totient
2012	Guts	TS2012	9	Combinatorics	Expected value	Expected value of permutation sorting moves
2012	Guts	TS2012	10	Combinatorics	Expected value	Expected heads after random coin flips
2012	Guts	TorS2012	1	Number Theory	Divisibility, primes & factorization	Factor 8051
2012	Guts	TorS2012	2	Algebra	Logarithms & exponents	Simplify logarithm expression
2012	Guts	TorS2012	3	Combinatorics	Counting & arrangements	mRNA sequences with no A or no U
2012	Guts	TorS2012	4	Number Theory	Modular arithmetic	Smallest n with 17^n+n divisible by 29
2012	Guts	TorS2012	5	Geometry	Triangles (lengths, angles, areas)	Right triangle inscribed rectangle area
2012	Guts	TorS2012	6	Combinatorics	Counting & arrangements	Count 3-faces in 5-cube
2012	Guts	TorS2012	7	Combinatorics	Expected value	Expected empty bins
2012	Guts	TorS2012	8	Number Theory	Modular arithmetic	Binomial coefficient mod 221
2012	Guts	TorS2012	9	Geometry	Similarity, ratios & mass points	Angle bisectors in right triangle, ratio BG/GF
2012	Guts	TorS2012	10	Number Theory	Divisibility, primes & factorization	n+7 divides n^2-2012
2012	Guts	TorS2012	11	Geometry	Polygons & area	Largest equilateral triangle in regular pentagon
2012	Guts	TorS2012	12	Combinatorics	Expected value	Coupon collector with 7 notes

Year	Test	Source file	#	Subject	Topic	Key idea
2012	Guts	TorS2012	13	Combinatorics	Expected value	Average of subset averages
2012	Guts	TorS2012	14	Algebra	Arithmetic & word problems	Clock hands right angles in a day
2012	Guts	TorS2012	15	Other	Logic & puzzles	Card flipping puzzle, equal face-up piles
2012	Guts	TorS2012	16	Algebra	Polynomials & Vieta	Cubic with consecutive integer roots, maximize ratio
2012	Guts	TorS2012	17	Combinatorics	Expected value	Expected blue balls Polya-like urn
2012	Guts	TorS2012	18	Geometry	Circles (tangents, power of a point, cyclic)	Intersecting circles, lengths via power of point
2012	Guts	TorS2012	19	Algebra	Sequences, series & recurrences	Harmonic sum identity
2012	Guts	TorS2012	20	Geometry	Circles (tangents, power of a point, cyclic)	Circles tangent to three mutually tangent circles
2012	Guts	TorS2012	21	Number Theory	Modular arithmetic	Sum over binary digit counts mod 629
2012	Guts	TorS2012	22	Geometry	Polygons & area	Parallelogram subdivided, area ratio
2012	Guts	TorS2012	23	Number Theory	Arithmetic functions & valuations	Totient maximal values, sum of n
2012	Guts	TorS2012	24	Combinatorics	Inclusion-exclusion, bijections & lattice paths	Lattice paths with exactly 6 turns
2012	Guts	TorS2012	25	Number Theory	Modular arithmetic	4^2012 mod n
2012	Guts	TorS2012	26	Geometry	Triangles (lengths, angles, areas)	Angles from unit squares side by side
2012	Guts	TorS2012	27	Algebra	Functions & functional equations	Express rational expression in terms of f(x)
2012	Guts	TorS2012	28	Geometry	Triangles (lengths, angles, areas)	Incenters in right triangle, area BGH
2012	Guts	TorS2012	29	Combinatorics	Expected value	Expected number of 6-inversions in permutation
2012	Guts	TorS2012	30	Combinatorics	Expected value	Expected sum of squared pile sizes
2012	Guts	TorS2012	31	Algebra	Equations & systems	Solve for n from 2^n+1, then consecutive squares
2012	Guts	TorS2012	32	Combinatorics	Expected value	Optimal stopping with rolling dice
2012	Guts	TorS2012	33	Algebra	Polynomials & Vieta	Degree-6 polynomial equal 7 at six points, coefficient
2012	Guts	TorS2012	34	Number Theory	Diophantine equations	20a+12b and 20b-12a powers of 2
2012	Guts	TorS2012	35	Geometry	Triangles (lengths, angles, areas)	Angle AEC in square and rhombus
2012	Guts	TorS2012	36	Combinatorics	Games, invariants & pigeonhole	Coin flipping blindfolded rotating card strategy
2012	Guts	TorS2012	37	Combinatorics	Counting & arrangements	Permutations with exactly one descent
2012	Guts	TorS2012	38	Combinatorics	Counting & arrangements	Integer triangles with perimeter 2012
2012	Guts	TorS2012	39	Number Theory	Arithmetic functions & valuations	Totient equation x/phi(x)=3 reciprocal sum
2012	Guts	TorS2012	40	Number Theory	Arithmetic functions & valuations	Sum of largest odd divisors
2012	Guts	TorS2012	41	Geometry	Polygons & area	Infinite squares inscribed in triangle, area sum
2012	Guts	TorS2012	42	Geometry	Coordinate & analytic geometry	Parabola focus chord tangents angle
2013	Calculus	AnaS2013	1	Algebra	Equations & systems	Linear system in a,b,c
2013	Calculus	AnaS2013	2	Algebra	Inequalities & optimization	Max distance to origin on x^4+y^4=1
2013	Calculus	AnaS2013	3	Calculus	Limits	Limit of sin2x over exponential difference
2013	Calculus	AnaS2013	4	Algebra	Complex numbers & roots of unity	Complex z with Im(z)=z^2-z
2013	Calculus	AnaS2013	5	Calculus	Series (calculus)	Abel summation of alternating series
2013	Calculus	AnaS2013	6	Algebra	Polynomials & Vieta	Minimal polynomial of cos(pi/7) evaluated at -1
2013	Calculus	AnaS2013	7	Algebra	Inequalities & optimization	Minimize y given cubic constraint
2013	Calculus	AnaS2013	8	Calculus	Applications (area, volume, rates)	Related rates, angle of sight

Year	Test	Source file	#	Subject	Topic	Key idea
2013	Calculus	AnaS2013	9	Calculus	Advanced / tricky integrals	Integral with inverse-function symmetry
2013	Calculus	AnaS2013	10	Algebra	Functions & functional equations	Iterated absolute value functions, sharp points limit
2013	Calculus	AnaS2013	11	Algebra	Sequences, series & recurrences	Alternating binomial sum positivity proof
2013	Calculus	AnaS2013	12	Algebra	Polynomials & Vieta	Polynomial real roots bounds proof
2013	Discrete	DisS2013	1	Number Theory	Modular arithmetic	CRT with remainders 6 mod 25 and 7 mod 9
2013	Discrete	DisS2013	2	Combinatorics	Probability	Three 4-sided dice, sum coprime to product
2013	Discrete	DisS2013	3	Combinatorics	Games, invariants & pigeonhole	Minimum steps removing equal coins from piles
2013	Discrete	DisS2013	4	Number Theory	Modular arithmetic	Iterated linear function mod 2011
2013	Discrete	DisS2013	5	Algebra	Complex numbers & roots of unity	Roots of x^2013=2^2013 with k-th root condition
2013	Discrete	DisS2013	6	Combinatorics	Probability	Coin flips head then two tails at flip 12
2013	Discrete	DisS2013	7	Number Theory	Diophantine equations	Frobenius-type semigroup containment
2013	Discrete	DisS2013	8	Combinatorics	Expected value	Expected value of nested random functions
2013	Discrete	DisS2013	9	Combinatorics	Graphs, grids, colorings & tilings	Ball passing in circle, minimal passes diameter
2013	Discrete	DisS2013	10	Number Theory	Divisibility, primes & factorization	Smallest n for permutation of given order
2013	Discrete	DisS2013	11	Combinatorics	Probability	Parity of coin sum probability proof
2013	Discrete	DisS2013	12	Number Theory	Arithmetic functions & valuations	p-adic valuation of p^p factorial
2013	General	IS2013	1	Algebra	Arithmetic & word problems	Escalator speed and steps
2013	General	IS2013	2	Geometry	Circles (tangents, power of a point, cyclic)	Perimeter of semicircle logo arcs
2013	General	IS2013	3	Combinatorics	Probability	Two methods of drawing red ball
2013	General	IS2013	4	Geometry	Circles (tangents, power of a point, cyclic)	Circle tangent to square and semicircle
2013	General	IS2013	5	Number Theory	Divisibility, primes & factorization	gcd and lcm of m,n with given relations
2013	General	IS2013	6	Geometry	Transformations, loci & advanced tools	Shortest path via x-axis, reflection
2013	General	IS2013	7	Algebra	Polynomials & Vieta	Symmetric sums a+b-c, ab-bc-ca, abc
2013	General	IS2013	8	Number Theory	Digits & bases	Prime with base 2 and base 5 representations
2013	General	IS2013	9	Algebra	Complex numbers & roots of unity	Ant spiral with 60 degree turns, convergence point
2013	General	IS2013	10	Combinatorics	Expected value	Expected same-colored edges in 3x3 board
2013	General	IS2013	11	Other	Linear algebra & misc	Iterating (a+b,b+c,c+a) returning permutation
2013	General	IS2013	12	Geometry	Triangles (lengths, angles, areas)	Triangle angles from logs, altitudes
2013	General	IS2013	13	Algebra	Functions & functional equations	Functional equation f(a)f(b)=f(a+b)+f(a-b)
2013	General	IS2013	14	Geometry	Triangles (lengths, angles, areas)	Incircle and midpoint, perpendicular to BO, find CE
2013	General	IS2013	15	Geometry	Similarity, ratios & mass points	Quadrilateral with equal angles, similar triangles
2013	General	IS2013	16	Algebra	Polynomials & Vieta	Integer-valued cubic polynomial divisors
2013	General	IS2013	17	Algebra	Sequences, series & recurrences	Linear recurrence averaging N previous terms limit
2013	General	IS2013	18	Combinatorics	Probability	Octahedron face arrows, probability visits all faces
2013	General	IS2013	19	Geometry	Circles (tangents, power of a point, cyclic)	Equilateral triangle inscribed, chord lengths integers
2013	General	IS2013	20	Algebra	Sequences, series & recurrences	Infinite product from repeated square roots
2013	Geometry	GeoS2013	1	Geometry	Polygons & area	Rectangle perimeter from area and diagonal
2013	Geometry	GeoS2013	2	Geometry	Circles (tangents, power of a point, cyclic)	Circle tangent to two rays and larger circle

Year	Test	Source file	#	Subject	Topic	Key idea
2013	Geometry	GeoS2013	3	Geometry	3D geometry	Angle between vertices of regular tetrahedron center
2013	Geometry	GeoS2013	4	Geometry	3D geometry	Two cubes volume and surface area relations
2013	Geometry	GeoS2013	5	Geometry	Transformations, loci & advanced tools	Locus of points with angle APB over 120 degrees
2013	Geometry	GeoS2013	6	Geometry	Circles (tangents, power of a point, cyclic)	Cyclic quadrilateral, extend sides, area of triangle
2013	Geometry	GeoS2013	7	Geometry	Coordinate & analytic geometry	Midpoint of PQ, range of linear expression
2013	Geometry	GeoS2013	8	Geometry	Triangles (lengths, angles, areas)	Incenter and excircle configuration, area EFP
2013	Geometry	GeoS2013	9	Geometry	Circles (tangents, power of a point, cyclic)	Two circumcircles meet at X, circumradius
2013	Geometry	GeoS2013	10	Geometry	Circles (tangents, power of a point, cyclic)	Incircle tangency, harmonic reciprocal lengths
2013	Geometry	GeoS2013	11	Geometry	Polygons & area	Convex polygon perimeter 1 covered by circle proof
2013	Geometry	GeoS2013	12	Geometry	Circles (tangents, power of a point, cyclic)	Tangents to circle, midpoint of chord angle proof
2013	Guts	TS2013	1	Algebra	Arithmetic & word problems	Reflexive clock times count
2013	Guts	TS2013	2	Algebra	Equations & systems	Cube roots sum is integer
2013	Guts	TS2013	3	Combinatorics	Counting & arrangements	Round-robin tournament placements
2013	Guts	TS2013	4	Algebra	Equations & systems	Radical equation x^2=5x+6sqrt(x-3)
2013	Guts	TS2013	5	Geometry	Circles (tangents, power of a point, cyclic)	Two circles, perpendicular, find chord AB
2013	Guts	TS2013	6	Combinatorics	Graphs, grids, colorings & tilings	Balanced triples in regular graph
2013	Guts	TS2013	7	Algebra	Sequences, series & recurrences	Fibonacci generating function equals 2
2013	Guts	TS2013	8	Geometry	Coordinate & analytic geometry	Parabola focal chord, triangle area
2013	Guts	TS2013	9	Algebra	Sequences, series & recurrences	Simultaneous sequences with square roots, sum of squares
2013	Guts	TS2013	10	Combinatorics	Graphs, grids, colorings & tilings	Counting spanning trees of connected cliques
2014	Calculus	AnaS2014	1	Algebra	Logarithms & exponents	Solve 4^x - 2^(x+2)+3=0
2014	Calculus	AnaS2014	2	Algebra	Polynomials & Vieta	Smallest root of cubic
2014	Calculus	AnaS2014	3	Algebra	Arithmetic & word problems	Distance markers passing train, range of x
2014	Calculus	AnaS2014	4	Algebra	Polynomials & Vieta	Quintic with roots in arithmetic progression
2014	Calculus	AnaS2014	5	Calculus	Limits	Limit of ratio of square roots at infinity
2014	Calculus	AnaS2014	6	Algebra	Functions & functional equations	Functional equation with x and 2/x
2014	Calculus	AnaS2014	7	Algebra	Polynomials & Vieta	Polynomial with sums of roots of two quadratics
2014	Calculus	AnaS2014	8	Algebra	Sequences, series & recurrences	Sum of f(n)/2^n from partial sums
2014	Calculus	AnaS2014	9	Calculus	Advanced / tricky integrals	Asymptotics of Laplace-type integral
2014	Calculus	AnaS2014	10	Algebra	Equations & systems	Approximate roots of perturbed cubic
2014	Calculus	AnaS2014	11	Algebra	Inequalities & optimization	Maximize a+b+c+d under constraints
2014	Calculus	AnaS2014	12	Algebra	Polynomials & Vieta	Repeated roots count for coprime polynomials proof
2014	General	IS2014	1	Combinatorics	Counting & arrangements	Four-digit numbers using digits 0,1,2,4
2014	General	IS2014	2	Geometry	Triangles (lengths, angles, areas)	Similar triangles with swapped sides, area
2014	General	IS2014	3	Algebra	Arithmetic & word problems	Boba and burger prices, coin count
2014	General	IS2014	4	Combinatorics	Counting & arrangements	Circular seating with restrictions
2014	General	IS2014	5	Combinatorics	Probability	Compare heads of 2014 and 2015 coins
2014	General	IS2014	6	Number Theory	Divisibility, primes & factorization	m+n and m-n primes, maximize mn

Year	Test	Source file	#	Subject	Topic	Key idea
2014	General	IS2014	7	Algebra	Equations & systems	Quadratic form f(a,b) system, find |a+b|
2014	General	IS2014	8	Geometry	Circles (tangents, power of a point, cyclic)	Tangent from B to small circle, triangle AMN
2014	General	IS2014	9	Algebra	Sequences, series & recurrences	Coupled linear recurrences a_n, b_n
2014	General	IS2014	10	Calculus	Applications (area, volume, rates)	Sphere cross-section area rate, plane distance
2014	General	IS2014	11	Combinatorics	Probability	Geometric probability obtuse triangle
2014	General	IS2014	12	Geometry	Triangles (lengths, angles, areas)	Maximal area of triangle ACD
2014	General	IS2014	13	Geometry	3D geometry	Cylinder inscribed in sphere, volume ratio
2014	General	IS2014	14	Algebra	Inequalities & optimization	Maximize sum of f(x_i) with fixed sum
2014	General	IS2014	15	Combinatorics	Probability	Extending match series probability
2014	General	IS2014	16	Number Theory	Digits & bases	Digit rotation multiple of 7
2014	General	IS2014	17	Geometry	3D geometry	Faces and edges of permutohedron
2014	General	IS2014	18	Number Theory	Arithmetic functions & valuations	Binomial basis coefficient, 2-adic valuation
2014	General	IS2014	19	Calculus	Integration techniques	Integral of sqrt(tan)
2014	General	IS2014	20	Geometry	Circles (tangents, power of a point, cyclic)	Three equal circles through point, enclosing circle
2014	Geometry	GeoS2014	1	Geometry	Polygons & area	Incircle area to hexagon area
2014	Geometry	GeoS2014	2	Geometry	Polygons & area	Regular hexagon quadrilateral area
2014	Geometry	GeoS2014	3	Geometry	Circles (tangents, power of a point, cyclic)	Isosceles triangle, circle on diameter BC
2014	Geometry	GeoS2014	4	Geometry	3D geometry	Spheres packed in cylinder
2014	Geometry	GeoS2014	5	Geometry	3D geometry	Hypersphere packing in 100-cube
2014	Geometry	GeoS2014	6	Geometry	Coordinate & analytic geometry	Square, arc, midpoint line intersection
2014	Geometry	GeoS2014	7	Geometry	Similarity, ratios & mass points	Parallelogram with BF=EG, ratios
2014	Geometry	GeoS2014	8	Geometry	Polygons & area	Region bounded by chords and arc in semicircle
2014	Geometry	GeoS2014	9	Geometry	Transformations, loci & advanced tools	Equilateral triangles on sides, Fermat-like point
2014	Geometry	GeoS2014	10	Combinatorics	Expected value	Expected area of random knight-move polygon
2014	Geometry	GeoS2014	11	Geometry	Triangles (lengths, angles, areas)	Exradii reciprocal identity proof
2014	Geometry	GeoS2014	12	Geometry	Transformations, loci & advanced tools	Circumcircle passes through fixed point proof
2014	Guts	TS2014	1	Combinatorics	Counting & arrangements	Sum of binomial coefficients 2^7
2014	Guts	TS2014	2	Combinatorics	Expected value	Expected books with symmetric letters
2014	Guts	TS2014	3	Number Theory	Diophantine equations	Integer money amounts from reciprocal equation
2014	Guts	TS2014	4	Geometry	Triangles (lengths, angles, areas)	Right triangle altitude segments, area and circumradius
2014	Guts	TS2014	5	Geometry	Polygons & area	Supplementary regular polygons
2014	Guts	TS2014	6	Calculus	Integration techniques	Integrate vertical velocity 1/(1-t^4)
2014	Guts	TS2014	7	Geometry	3D geometry	Cube and pyramid equal-volume excess
2014	Guts	TS2014	8	Combinatorics	Counting & arrangements	Binomial coefficients in combos of books
2014	Guts	TS2014	9	Calculus	Limits	Products of functions with finite limit
2014	Guts	TS2014	10	Number Theory	Arithmetic functions & valuations	Sum of unitary divisors
2014	Guts	TS2014	11	Algebra	Polynomials & Vieta	Reduce x^100 modulo x^10+x+1
2014	Guts	TS2014	12	Number Theory	Divisibility, primes & factorization	Reversible numbers with equal divisor counts

Year	Test	Source file	#	Subject	Topic	Key idea
2014	Guts	TS2014	13	Geometry	Similarity, ratios & mass points	Cevians intersect, area of quadrilateral
2014	Guts	TS2014	14	Geometry	Coordinate & analytic geometry	Intersections of parabola and ellipse, sum of x
2014	Guts	TS2014	15	Combinatorics	Probability	Order of color exhaustion in drawing balls
2014	Guts	TS2014	16	Calculus	Series (calculus)	Double sum with min(n,k) geometric weights
2014	Guts	TS2014	17	Combinatorics	Expected value	Random walk with reflecting walls, expected time
2014	Guts	TS2014	18	Combinatorics	Probability	Monty Hall variant with five boxes
2014	Guts	TS2014	19	Number Theory	Digits & bases	Cyclic numbers in base 18
2014	Guts	TS2014	20	Calculus	Advanced / tricky integrals	Bessel integral against Gaussian
2015	Calculus	AnaS2015	1	Algebra	Logarithms & exponents	Sum of powers of two equals 24.375, product
2015	Calculus	AnaS2015	2	Algebra	Sequences, series & recurrences	Generating function coefficient
2015	Calculus	AnaS2015	3	Algebra	Equations & systems	Integer solutions to quadratic system
2015	Calculus	AnaS2015	4	Algebra	Sequences, series & recurrences	Linear recurrence, sum of terms
2015	Calculus	AnaS2015	5	Algebra	Inequalities & optimization	Max and min of x^2+y^2 on circle
2015	Calculus	AnaS2015	6	Algebra	Polynomials & Vieta	Geometric progression roots, reciprocal sum
2015	Calculus	AnaS2015	7	Algebra	Complex numbers & roots of unity	Alternating binomial sum via complex numbers
2015	Calculus	AnaS2015	8	Algebra	Complex numbers & roots of unity	Product over 7th roots of unity
2015	Calculus	AnaS2015	9	Calculus	Limits	Riemann sum limit of sqrt(n^2-k^2)
2015	Calculus	AnaS2015	10	Calculus	Advanced / tricky integrals	Integral of ln(4 sin x)
2015	Calculus	AnaS2015	11	Algebra	Complex numbers & roots of unity	Sum of distances to roots of unity proof
2015	Calculus	AnaS2015	12	Algebra	Polynomials & Vieta	Count monic divisors of irreducible polynomial
2015	General	IS2015	1	Combinatorics	Counting & arrangements	Tea sequences no repeats on successive days
2015	General	IS2015	2	Number Theory	Digits & bases	2015th digit in list of even numbers
2015	General	IS2015	3	Geometry	Triangles (lengths, angles, areas)	Quadrilateral with right angle, BD squared
2015	General	IS2015	4	Algebra	Arithmetic & word problems	Train, bridge, and walking man
2015	General	IS2015	5	Combinatorics	Counting & arrangements	Partition of 10-set into two parts
2015	General	IS2015	6	Algebra	Functions & functional equations	Multiplicative increasing function, f(2015)
2015	General	IS2015	7	Geometry	Triangles (lengths, angles, areas)	Contact triangle angles from incircle
2015	General	IS2015	8	Combinatorics	Probability	Digit sum 19 probability
2015	General	IS2015	9	Number Theory	Modular arithmetic	Missing digit of 2^29 via mod 9
2015	General	IS2015	10	Combinatorics	Counting & arrangements	Nesting boxes counting
2015	General	IS2015	11	Algebra	Polynomials & Vieta	Sum of cubes of pairwise root sums
2015	General	IS2015	12	Combinatorics	Counting & arrangements	Maximum non-attacking bishops arrangements
2015	General	IS2015	13	Combinatorics	Expected value	Expected white squares surrounded by rooks
2015	General	IS2015	14	Algebra	Complex numbers & roots of unity	Product of complex expression over k
2015	General	IS2015	15	Geometry	3D geometry	Rotational symmetries of icosahedron
2015	General	IS2015	16	Combinatorics	Counting & arrangements	Binary decision trees, power of 2 dividing
2015	General	IS2015	17	Geometry	Circles (tangents, power of a point, cyclic)	Circle and square union area
2015	General	IS2015	18	Calculus	Series (calculus)	Telescoping sum 1/((2n-1)(3n-1))

Year	Test	Source file	#	Subject	Topic	Key idea
2015	General	IS2015	19	Combinatorics	Probability	Chain of truthfulness statements probability
2015	General	IS2015	20	Algebra	Inequalities & optimization	Palindromic quartic real root, minimize 8a^2+b^2
2015	Geometry	GeoS2015	1	Geometry	Triangles (lengths, angles, areas)	Angle bisectors incenter angle BDC
2015	Geometry	GeoS2015	2	Geometry	Polygons & area	Overlap of two triangles in hexagon
2015	Geometry	GeoS2015	3	Geometry	Triangles (lengths, angles, areas)	Maximal triangle area given cevian pieces
2015	Geometry	GeoS2015	4	Geometry	Circles (tangents, power of a point, cyclic)	Two tangent circles in rectangle corners
2015	Geometry	GeoS2015	5	Geometry	Coordinate & analytic geometry	Parallelogram fourth vertices, triangle area
2015	Geometry	GeoS2015	6	Geometry	3D geometry	Line through sphere, second intersection length
2015	Geometry	GeoS2015	7	Geometry	3D geometry	Minimize sum of distances reflecting across plane
2015	Geometry	GeoS2015	8	Geometry	Coordinate & analytic geometry	Maximal area inscribed quadrilateral in ellipse
2015	Geometry	GeoS2015	9	Geometry	Circles (tangents, power of a point, cyclic)	Circle on diameter BC, harmonic ratio EX/XF
2015	Geometry	GeoS2015	10	Geometry	Circles (tangents, power of a point, cyclic)	Circle through E,F tangent to BC, ratio of squares
2015	Geometry	GeoS2015	11	Geometry	Circles (tangents, power of a point, cyclic)	Two intersecting circles, equal segments proof
2015	Geometry	GeoS2015	12	Geometry	Transformations, loci & advanced tools	Locus of centers of tangent circles is parabola
2015	Guts	TS2015	1	Combinatorics	Expected value	Expected product of rolls until stop
2015	Guts	TS2015	2	Number Theory	Digits & bases	Digit sum of 1001^10
2015	Guts	TS2015	3	Combinatorics	Counting & arrangements	Grid placement with coprime neighbors
2015	Guts	TS2015	4	Geometry	Similarity, ratios & mass points	Parallel lines splitting triangle area, inner triangle
2015	Guts	TS2015	5	Number Theory	Divisibility, primes & factorization	Smallest 0-1 number divisible by 1 through 9
2015	Guts	TS2015	6	Combinatorics	Counting & arrangements	Chains of strictly nested subsets
2015	Guts	TS2015	7	Geometry	Transformations, loci & advanced tools	Equal segments and angles polygon, possible X3
2015	Guts	TS2015	8	Algebra	Sequences, series & recurrences	Limit of recursive cube-root sequence
2015	Guts	TS2015	9	Geometry	3D geometry	Largest square inscribed in unit cube
2015	Guts	TS2015	10	Algebra	Polynomials & Vieta	Quadratics roots in arithmetic progression
2015	Guts	TS2015	11	Combinatorics	Probability	Random adjacent merging probability
2015	Guts	TS2015	12	Combinatorics	Inclusion-exclusion, bijections & lattice paths	Counting pairs bijection, generating-series difference
2015	Guts	TS2015	13	Number Theory	Diophantine equations	Pythagorean triples with legs differing by 1
2015	Guts	TS2015	14	Combinatorics	Inclusion-exclusion, bijections & lattice paths	Lattice paths meeting probability
2015	Guts	TS2015	15	Calculus	Integration techniques	Integral using x to 1/x symmetry
2015	Guts	TS2015	16	Geometry	3D geometry	Skew pentagon with right angles, cosine
2015	Guts	TS2015	17	Algebra	Inequalities & optimization	Cyclic inequality constants minimal x/y
2015	Guts	TS2015	18	Geometry	Transformations, loci & advanced tools	Region of friendly points between lines
2015	Guts	TS2015	19	Algebra	Sequences, series & recurrences	Recursive sequences ratio limit
2015	Guts	TS2015	20	Combinatorics	Expected value	Random Tower of Hanoi expected moves
2016	Calculus	AnaS2016	1	Algebra	Arithmetic & word problems	Difference of cubes
2016	Calculus	AnaS2016	2	Algebra	Logarithms & exponents	System with log base 2 of powers
2016	Calculus	AnaS2016	3	Algebra	Arithmetic & word problems	Train, tunnel and running Steve
2016	Calculus	AnaS2016	4	Algebra	Sequences, series & recurrences	Geometric progression with odd-even difference

Year	Test	Source file	#	Subject	Topic	Key idea
2016	Calculus	AnaS2016	5	Algebra	Trigonometry (algebraic)	Sum of tan/(1+tan) over 1..89 degrees
2016	Calculus	AnaS2016	6	Calculus	Applications (area, volume, rates)	Implicit curve velocity components
2016	Calculus	AnaS2016	7	Algebra	Polynomials & Vieta	Coefficient of x^2 in product polynomial
2016	Calculus	AnaS2016	8	Calculus	Limits	Limit of power series to 1/x
2016	Calculus	AnaS2016	9	Algebra	Polynomials & Vieta	Polynomial from second derivative, root power sums
2016	Calculus	AnaS2016	10	Calculus	Series (calculus)	Series with central binomial times power
2016	Discrete	DisS2016	1	Algebra	Arithmetic & word problems	Average coin value, least nickels
2016	Discrete	DisS2016	2	Combinatorics	Inclusion-exclusion, bijections & lattice paths	Count divisible by 3 or 7 not 21
2016	Discrete	DisS2016	3	Combinatorics	Counting & arrangements	Full house poker hands
2016	Discrete	DisS2016	4	Combinatorics	Inclusion-exclusion, bijections & lattice paths	Derangement of socks among three people
2016	Discrete	DisS2016	5	Number Theory	Modular arithmetic	Last two digits of power tower
2016	Discrete	DisS2016	6	Combinatorics	Games, invariants & pigeonhole	Maximize repeated replacement 2x+y
2016	Discrete	DisS2016	7	Combinatorics	Graphs, grids, colorings & tilings	Chromatic number of divisibility graph
2016	Discrete	DisS2016	8	Other	Linear algebra & misc	Hypercube quadratic form minimum, eigenvalues
2016	Discrete	DisS2016	9	Number Theory	Modular arithmetic	Conjugate powers, last two digits
2016	Discrete	DisS2016	10	Combinatorics	Graphs, grids, colorings & tilings	Domino tilings with no fault line
2016	General	IS2016	1	Combinatorics	Inclusion-exclusion, bijections & lattice paths	Sum of integers divisible by 2, 3 or 5
2016	General	IS2016	2	Algebra	Arithmetic & word problems	Custom binary operation evaluation
2016	General	IS2016	3	Geometry	Polygons & area	Perimeters of equilateral triangle and square
2016	General	IS2016	4	Geometry	Triangles (lengths, angles, areas)	Sum of altitudes from interior point
2016	General	IS2016	5	Number Theory	Diophantine equations	Gaussian square equals integer plus 46i
2016	General	IS2016	6	Number Theory	Modular arithmetic	Recurrence divisible by 5 count
2016	General	IS2016	7	Algebra	Inequalities & optimization	Minimize sum of cubes with x+y=0
2016	General	IS2016	8	Algebra	Logarithms & exponents	Denest cube-root sum rationalize
2016	General	IS2016	9	Combinatorics	Expected value	Monochromatic triangles in colored lattice
2016	General	IS2016	10	Calculus	Limits	Riemann-type limit of sum i(n+1-i)
2016	General	IS2016	11	Algebra	Polynomials & Vieta	Cubic roots in arithmetic progression
2016	General	IS2016	12	Combinatorics	Counting & arrangements	Nondecreasing sequences by stars and bars
2016	General	IS2016	13	Algebra	Polynomials & Vieta	Quartic and cubic through common points, fourth intersection
2016	General	IS2016	14	Geometry	Circles (tangents, power of a point, cyclic)	Three unit circles in square, minimal side
2016	General	IS2016	15	Combinatorics	Counting & arrangements	Pairwise-intersecting-in-one 3-subsets of 6-set
2016	General	IS2016	16	Geometry	3D geometry	Inscribed sphere of corner tetrahedron
2016	General	IS2016	17	Geometry	Triangles (lengths, angles, areas)	30-60-90 triangle with angle bisector, quadrilateral area
2016	General	IS2016	18	Calculus	Limits	Average of sum of two squares counts
2016	General	IS2016	19	Geometry	3D geometry	Centroid iteration of tetrahedra limit point
2016	General	IS2016	20	Algebra	Complex numbers & roots of unity	Product of roots of unity and cosines
2016	Geometry	GeoS2016	1	Geometry	3D geometry	Prism and cube surface areas
2016	Geometry	GeoS2016	2	Geometry	Circles (tangents, power of a point, cyclic)	Cyclic quadrilateral area

Year	Test	Source file	#	Subject	Topic	Key idea
2016	Geometry	GeoS2016	3	Geometry	Triangles (lengths, angles, areas)	Probability integer triangle acute
2016	Geometry	GeoS2016	4	Geometry	Triangles (lengths, angles, areas)	Square inscribed in equilateral triangle area ratio
2016	Geometry	GeoS2016	5	Geometry	Circles (tangents, power of a point, cyclic)	Angle chasing in pentagon
2016	Geometry	GeoS2016	6	Geometry	Triangles (lengths, angles, areas)	Triangle formed by medians lengths, area
2016	Geometry	GeoS2016	7	Geometry	Triangles (lengths, angles, areas)	Right isosceles triangle with 30-60 triangle area
2016	Geometry	GeoS2016	8	Geometry	3D geometry	Circumradius of regular 7-simplex
2016	Geometry	GeoS2016	9	Geometry	Triangles (lengths, angles, areas)	Squares on right triangle sides, lengths
2016	Geometry	GeoS2016	10	Geometry	Circles (tangents, power of a point, cyclic)	Chain of tangent circles to incircle, area sum
2016	Guts	TS2016	1	Algebra	Sequences, series & recurrences	Recurrence a^2-2 with sqrt3 start
2016	Guts	TS2016	2	Geometry	Polygons & area	Octagon cut from unit square
2016	Guts	TS2016	3	Algebra	Arithmetic & word problems	Mobius strip train travel time
2016	Guts	TS2016	4	Combinatorics	Graphs, grids, colorings & tilings	Count graphs with given degree sequence
2016	Guts	TS2016	5	Geometry	Triangles (lengths, angles, areas)	Right isosceles plus 30-60 triangle area
2016	Guts	TS2016	6	Number Theory	Divisibility, primes & factorization	Integers below 400 with three square divisors
2016	Guts	TS2016	7	Algebra	Functions & functional equations	Discrete harmonic function, determining points
2016	Guts	TS2016	8	Combinatorics	Counting & arrangements	Candies distribution Berkeley and Stanford
2016	Guts	TS2016	9	Combinatorics	Recursion & states	Subsets without three consecutive integers
2016	Guts	TS2016	10	Geometry	Coordinate & analytic geometry	Lattice triangle area 1/2 minimal perimeter
2016	Guts	TS2016	11	Geometry	Circles (tangents, power of a point, cyclic)	Two circles, tangent line, power of point ratio
2016	Guts	TS2016	12	Calculus	Applications (area, volume, rates)	Center of mass of hemisphere
2016	Guts	TS2016	13	Combinatorics	Expected value	Expected turns for random negative-binomial count
2016	Guts	TS2016	14	Combinatorics	Counting & arrangements	Max shattered points by axis-aligned boxes
2016	Guts	TS2016	15	Algebra	Polynomials & Vieta	Product over roots of quartic expression
2017	Calculus	AnaS2017	1	Algebra	Inequalities & optimization	max students with high score given average
2017	Calculus	AnaS2017	2	Algebra	Logarithms & exponents	exponential equation, substitute 3^x
2017	Calculus	AnaS2017	3	Calculus	Integration techniques	odd function integrates to zero
2017	Calculus	AnaS2017	4	Calculus	Series (calculus)	sum n^2/2^n
2017	Calculus	AnaS2017	5	Algebra	Floor, ceiling & absolute value	nested absolute value, three solutions
2017	Calculus	AnaS2017	6	Algebra	Polynomials & Vieta	Vieta, iterated linear map on roots
2017	Calculus	AnaS2017	7	Calculus	Series (calculus)	telescoping series
2017	Calculus	AnaS2017	8	Calculus	Advanced / tricky integrals	integral with floor, beta-type
2017	Calculus	AnaS2017	9	Algebra	Sequences, series & recurrences	generating function of congruent pairs, limit
2017	Calculus	AnaS2017	10	Algebra	Sequences, series & recurrences	binomial sum with harmonic numbers
2017	Discrete	DisS2017	1	Combinatorics	Probability	same-color socks
2017	Discrete	DisS2017	2	Combinatorics	Counting & arrangements	surjections onto 3 rounds
2017	Discrete	DisS2017	3	Number Theory	Arithmetic functions & valuations	smallest n with 7 proper divisors
2017	Discrete	DisS2017	4	Number Theory	Digits & bases	greatest multiple of 9 from digits
2017	Discrete	DisS2017	5	Combinatorics	Recursion & states	subsets without adjacent elements

Year	Test	Source file	#	Subject	Topic	Key idea
2017	Discrete	DisS2017	6	Combinatorics	Counting & arrangements	functions with f^5 constant 1, functional graph
2017	Discrete	DisS2017	7	Combinatorics	Expected value	expected lights with all neighbors on
2017	Discrete	DisS2017	8	Combinatorics	Probability	weak player wins elimination bracket
2017	Discrete	DisS2017	9	Combinatorics	Expected value	fourth moment of balls in box, limit
2017	Discrete	DisS2017	10	Number Theory	Arithmetic functions & valuations	alternating phi(n) binomial sum
2017	General	IS2017	1	Algebra	Arithmetic & word problems	age and height word problem
2017	General	IS2017	2	Geometry	Polygons & area	areas of equilateral triangle and square
2017	General	IS2017	3	Other	Logic & puzzles	letters with line symmetry
2017	General	IS2017	4	Number Theory	Diophantine equations	3-digit numbers ending 99 as n(n+2)
2017	General	IS2017	5	Algebra	Logarithms & exponents	log_a 16 = b integer pairs
2017	General	IS2017	6	Number Theory	Divisibility, primes & factorization	n with 2017 remainder 1 or 2
2017	General	IS2017	7	Algebra	Sequences, series & recurrences	infinite geometric series sum
2017	General	IS2017	8	Algebra	Equations & systems	x^3+y^3 from x+y and xy
2017	General	IS2017	9	Combinatorics	Counting & arrangements	sum of all permutations of digits
2017	General	IS2017	10	Geometry	Coordinate & analytic geometry	parallelogram fourth vertices, hull area
2017	General	IS2017	11	Combinatorics	Counting & arrangements	max handshakes, complete multipartite graph
2017	General	IS2017	12	Combinatorics	Expected value	average floor of rectangle perimeter, random point
2017	General	IS2017	13	Algebra	Inequalities & optimization	maximize tangent triangle area over r
2017	General	IS2017	14	Number Theory	Modular arithmetic	first term of 31,331,... divisible by 17
2017	General	IS2017	15	Combinatorics	Expected value	expected meeting time of walkers on cube
2017	General	IS2017	16	Geometry	Similarity, ratios & mass points	centroid triangle is 1/9-scaled, maximize area
2017	General	IS2017	17	Geometry	Circles (tangents, power of a point, cyclic)	angle chase with circumcircle, diameter AO
2017	General	IS2017	18	Algebra	Sequences, series & recurrences	recurrence k_n/n! limit
2017	General	IS2017	19	Geometry	Coordinate & analytic geometry	inscribed Steiner ellipse center distance
2017	General	IS2017	20	Algebra	Complex numbers & roots of unity	roots of unity filter on cos binomial sum mod 17
2017	Guts	TS2017	1	Algebra	Sequences, series & recurrences	robot speeds triangular distance, race length
2017	Guts	TS2017	2	Number Theory	Divisibility, primes & factorization	number of divisors of 900
2017	Guts	TS2017	3	Algebra	Equations & systems	pairwise sums of four numbers
2017	Guts	TS2017	4	Combinatorics	Probability	two random darts, acute angle
2017	Guts	TS2017	5	Combinatorics	Recursion & states	elevator stops, no adjacent floors
2017	Guts	TS2017	6	Combinatorics	Probability	random square center inside circle
2017	Guts	TS2017	7	Algebra	Inequalities & optimization	minimize x+y+z with xyz=86400
2017	Guts	TS2017	8	Algebra	Functions & functional equations	recursive sum f(n) from conditions
2017	Guts	TS2017	9	Geometry	Circles (tangents, power of a point, cyclic)	diameter, incircle, length BD
2017	Guts	TS2017	10	Combinatorics	Games, invariants & pigeonhole	guaranteed hit on 7x7 grid line
2017	Guts	TS2017	11	Number Theory	Modular arithmetic	probability n is primitive root mod 2017
2017	Guts	TS2017	12	Combinatorics	Probability	random walk returns to origin
2017	Guts	TS2017	13	Geometry	Polygons & area	common area of four triangles in square

Year	Test	Source file	#	Subject	Topic	Key idea
2017	Guts	TS2017	14	Algebra	Inequalities & optimization	max x+1/x given sum and reciprocal sum
2017	Guts	TS2017	15	Geometry	Transformations, loci & advanced tools	rotation about equilateral triangle, distance CP
2018	Algebra	ALSP18	1	Algebra	Arithmetic & word problems	runway length from acceleration, deceleration
2018	Algebra	ALSP18	2	Algebra	Polynomials & Vieta	cubic with a single distinct root
2018	Algebra	ALSP18	3	Algebra	Polynomials & Vieta	minimum degree via finite differences
2018	Algebra	ALSP18	4	Number Theory	Modular arithmetic	sum i(i+1)(i+2) mod 2018
2018	Algebra	ALSP18	5	Algebra	Polynomials & Vieta	cubic and quadratic sharing a root
2018	Algebra	ALSP18	6	Algebra	Inequalities & optimization	minimize sum of squares under constraint
2018	Algebra	ALSP18	7	Algebra	Sequences, series & recurrences	binomial sum over n!, exponential series
2018	Algebra	ALSP18	8	Algebra	Sequences, series & recurrences	alternating binomial sum with powers of 2
2018	Algebra	ALSP18	9	Algebra	Equations & systems	rational equation, telescoping-like factors
2018	Algebra	ALSP18	10	Algebra	Polynomials & Vieta	Newton power sums of cubic roots
2018	Calculus	CASP18	1	Calculus	Limits	nested radical integer part
2018	Calculus	CASP18	2	Calculus	Limits	AM-HM iteration limit
2018	Calculus	CASP18	3	Calculus	Derivatives & optimization	100th derivative of e^x cos x
2018	Calculus	CASP18	4	Calculus	Limits	limit of integral with n x^n
2018	Calculus	CASP18	5	Calculus	Advanced / tricky integrals	symmetry trick integral with e^-x
2018	Calculus	CASP18	6	Calculus	Series (calculus)	product of cosines then geometric series
2018	Calculus	CASP18	7	Calculus	Limits	trig limit small-angle
2018	Calculus	CASP18	8	Calculus	Derivatives & optimization	maximal circle tangent to Gaussian curves
2018	Calculus	CASP18	9	Calculus	Integration techniques	improper integral partial fractions
2018	Calculus	CASP18	10	Calculus	Series (calculus)	Chebyshev polynomials integrated and summed
2018	Discrete	COSP18	1	Combinatorics	Counting & arrangements	assign inks to pens
2018	Discrete	COSP18	2	Combinatorics	Counting & arrangements	multisets of subject pairs for team
2018	Discrete	COSP18	3	Combinatorics	Counting & arrangements	rectangles avoiding a center point
2018	Discrete	COSP18	4	Combinatorics	Probability	string ending in 2018 within 9 digits
2018	Discrete	COSP18	5	Combinatorics	Games, invariants & pigeonhole	edge removal game on complete graph
2018	Discrete	COSP18	6	Calculus	Series (calculus)	double sum binomial over 3^(i+j)
2018	Discrete	COSP18	7	Combinatorics	Counting & arrangements	intersecting chords in 21-gon
2018	Discrete	COSP18	8	Combinatorics	Expected value	bottle passing around circle, last person distance
2018	Discrete	COSP18	9	Combinatorics	Counting & arrangements	permutations with power equal to reversal
2018	Discrete	COSP18	10	Combinatorics	Recursion & states	random 2xN grid tiling probability
2018	Discrete	NTSP18	1	Number Theory	Arithmetic functions & valuations	divisors of 17! that are multiples of 20
2018	Discrete	NTSP18	2	Number Theory	Divisibility, primes & factorization	p/q+q/p=17, divisibility of power sums
2018	Discrete	NTSP18	3	Combinatorics	Games, invariants & pigeonhole	subset sum divisible by 7, pigeonhole
2018	Discrete	NTSP18	4	Number Theory	Modular arithmetic	repeated 2018 number mod 15
2018	Discrete	NTSP18	5	Number Theory	Modular arithmetic	polynomial roots mod 31, inverse sum
2018	Discrete	NTSP18	6	Number Theory	Modular arithmetic	linear congruential generator sum mod 31

Year	Test	Source file	#	Subject	Topic	Key idea
2018	Discrete	NTSP18	7	Number Theory	Modular arithmetic	counting triples with quadratic form mod 11
2018	Discrete	NTSP18	8	Number Theory	Modular arithmetic	triangular numbers permutation mod n
2018	Discrete	NTSP18	9	Number Theory	Modular arithmetic	sum of inverses of x^2+1 mod 199
2018	Discrete	NTSP18	10	Algebra	Complex numbers & roots of unity	product with cos and roots of unity
2018	Geometry	GEOSP18	1	Geometry	3D geometry	cube surface area and volume ratio
2018	Geometry	GEOSP18	2	Geometry	Polygons & area	square inscribed in circle, area outside circle
2018	Geometry	GEOSP18	3	Geometry	Triangles (lengths, angles, areas)	max area of triangle with fixed perimeter
2018	Geometry	GEOSP18	4	Combinatorics	Counting & arrangements	six lines intersection points minimum
2018	Geometry	GEOSP18	5	Combinatorics	Probability	random point in square, obtuse triangles
2018	Geometry	GEOSP18	6	Geometry	Triangles (lengths, angles, areas)	integer triangle maximum area
2018	Geometry	GEOSP18	7	Geometry	Coordinate & analytic geometry	tangent from point to ellipse
2018	Geometry	GEOSP18	8	Geometry	Triangles (lengths, angles, areas)	triangle with inradius and longest side
2018	Geometry	GEOSP18	9	Geometry	Triangles (lengths, angles, areas)	Heronian triangle with consecutive sides
2018	Geometry	GEOSP18	10	Geometry	3D geometry	spherical cap area ratio, plane section
2018	Guts	TESP18	1	Geometry	Triangles (lengths, angles, areas)	right triangle with inradius 5, hypotenuse 34
2018	Guts	TESP18	2	Algebra	Logarithms & exponents	20^x 18^x = 2018^x number of solutions
2018	Guts	TESP18	3	Combinatorics	Probability	ordering probability in random line
2018	Guts	TESP18	4	Combinatorics	Graphs, grids, colorings & tilings	non-attacking bishops on chessboard
2018	Guts	TESP18	5	Combinatorics	Games, invariants & pigeonhole	signed sum parity, achievable integers
2018	Guts	TESP18	6	Combinatorics	Probability	painted unit cubes in 20x1x7 prism
2018	Guts	TESP18	7	Combinatorics	Games, invariants & pigeonhole	gossip problem, minimum calls
2018	Guts	TESP18	8	Combinatorics	Games, invariants & pigeonhole	ball moving game, invariant for winning
2018	Guts	TESP18	9	Geometry	Circles (tangents, power of a point, cyclic)	tangent circles, triangle PQR area
2018	Guts	TESP18	10	Geometry	3D geometry	box face diagonals triangle area
2018	Guts	TESP18	11	Other	Logic & puzzles	prime digits knowledge puzzle
2018	Guts	TESP18	12	Algebra	Functions & functional equations	functional equations determine f at 12/13
2018	Guts	TESP18	13	Algebra	Sequences, series & recurrences	sum n^2 over 2^(n/2)
2018	Guts	TESP18	14	Algebra	Sequences, series & recurrences	Fibonacci partial sums over 3^n
2018	Guts	TESP18	15	Geometry	Circles (tangents, power of a point, cyclic)	incircle, circumcircle radical axis, minimal AL
2019	Algebra	algebra-problems	1	Algebra	Inequalities & optimization	integers satisfying quadratic inequality
2019	Algebra	algebra-problems	2	Geometry	Coordinate & analytic geometry	reflection across y=x distance
2019	Algebra	algebra-problems	3	Number Theory	Diophantine equations	integer points on circle
2019	Algebra	algebra-problems	4	Combinatorics	Probability	geometric probability of disk in square
2019	Algebra	algebra-problems	5	Algebra	Equations & systems	exponent equals one cases
2019	Algebra	algebra-problems	6	Algebra	Inequalities & optimization	maximize xy on circle
2019	Algebra	algebra-problems	7	Algebra	Polynomials & Vieta	Vieta, sum of cubes of roots zero
2019	Algebra	algebra-problems	8	Algebra	Complex numbers & roots of unity	roots of unity filter for coin flips
2019	Algebra	algebra-problems	9	Algebra	Complex numbers & roots of unity	product of first-quadrant roots of unity

Year	Test	Source file	#	Subject	Topic	Key idea
2019	Algebra	algebra-problems	10	Number Theory	Diophantine equations	positive definite quadratic form, lattice count
2019	Discrete	discrete-problems	1	Combinatorics	Probability	runs of 2019 equal flips
2019	Discrete	discrete-problems	2	Combinatorics	Probability	Bayes with two-tailed coin
2019	Discrete	discrete-problems	3	Combinatorics	Counting & arrangements	handshake lemma, count hugs
2019	Discrete	discrete-problems	4	Number Theory	Diophantine equations	1/a+1/b = 35/2 pairs
2019	Discrete	discrete-problems	5	Number Theory	Modular arithmetic	2^1110 mod 1111
2019	Discrete	discrete-problems	6	Number Theory	Divisibility, primes & factorization	triangular numbers that are semiprimes
2019	Discrete	discrete-problems	7	Combinatorics	Counting & arrangements	average flux of digit strings
2019	Discrete	discrete-problems	8	Number Theory	Arithmetic functions & valuations	sum of n with phi(n)=20
2019	Discrete	discrete-problems	9	Algebra	Complex numbers & roots of unity	sum over roots of unity of 1/(1-z w)
2019	Discrete	discrete-problems	10	Number Theory	Arithmetic functions & valuations	sum of squares of coprime residues
2019	General	individual-problems	1	Algebra	Polynomials & Vieta	cubic with max at 1, find p(10)
2019	General	individual-problems	2	Geometry	Triangles (lengths, angles, areas)	circles intersecting, angle PAC
2019	General	individual-problems	3	Algebra	Functions & functional equations	f(x+y)=f(xy) implies constant
2019	General	individual-problems	4	Combinatorics	Probability	three chips none of type A
2019	General	individual-problems	5	Geometry	3D geometry	region on plane with angle constraints
2019	General	individual-problems	6	Geometry	3D geometry	paint area on die minus dots
2019	General	individual-problems	7	Geometry	Polygons & area	equilateral triangle truncated corners, area ratio
2019	General	individual-problems	8	Algebra	Floor, ceiling & absolute value	rounding of n/2019 density, expected value
2019	General	individual-problems	9	Combinatorics	Counting & arrangements	count almost-palindromes under 4 letters
2019	General	individual-problems	10	Geometry	Triangles (lengths, angles, areas)	square with angle ratio 3.5, area
2019	General	individual-problems	11	Geometry	Transformations, loci & advanced tools	chord rotating about vertices, path length of P
2019	General	individual-problems	12	Number Theory	Digits & bases	repunit squared with digit sum 85
2019	General	individual-problems	13	Geometry	Circles (tangents, power of a point, cyclic)	two circles with arcs, quadrilateral area
2019	General	individual-problems	14	Number Theory	Diophantine equations	clock times with square hhmm and minutes
2019	General	individual-problems	15	Combinatorics	Counting & arrangements	distinct integers from digits of 04072019
2019	General	individual-problems	16	Geometry	Similarity, ratios & mass points	hexagon from parallels through point, minimum area
2019	General	individual-problems	17	Geometry	Circles (tangents, power of a point, cyclic)	circle region bounded by chord and arc
2019	General	individual-problems	18	Combinatorics	Graphs, grids, colorings & tilings	chain of divisibility steps, maximize score
2019	General	individual-problems	19	Algebra	Inequalities & optimization	Chebyshev minimax approximation of x^3
2019	General	individual-problems	20	Algebra	Sequences, series & recurrences	Fibonacci-like generating sum, unit fractions
2019	Geometry	geometry-problems	1	Geometry	Polygons & area	area of polygon in triangular grid
2019	Geometry	geometry-problems	2	Geometry	Transformations, loci & advanced tools	largest set with no obtuse triangle
2019	Geometry	geometry-problems	3	Geometry	Triangles (lengths, angles, areas)	parallelogram midpoint, length NC
2019	Geometry	geometry-problems	4	Geometry	Triangles (lengths, angles, areas)	right triangle perimeter from area, hypotenuse
2019	Geometry	geometry-problems	5	Geometry	Coordinate & analytic geometry	ellipse region in complex plane, area
2019	Geometry	geometry-problems	6	Geometry	Similarity, ratios & mass points	angle bisector ratio in 3-4-5 triangle
2019	Geometry	geometry-problems	7	Geometry	Transformations, loci & advanced tools	locus of E for angle AGB over 30 degrees

Year	Test	Source file	#	Subject	Topic	Key idea
2019	Geometry	geometry-problems	8	Geometry	Transformations, loci & advanced tools	reflected centroids, circumcenter, length XY
2019	Geometry	geometry-problems	9	Geometry	3D geometry	tetrahedron with tangential quadrilaterals
2019	Geometry	geometry-problems	10	Geometry	Transformations, loci & advanced tools	3-4-5 point of isosceles triangle
2019	Guts	team-problems	1	Algebra	Complex numbers & roots of unity	maximize chord length on unit circle
2019	Guts	team-problems	2	Number Theory	Modular arithmetic	2^2019 mod 7
2019	Guts	team-problems	3	Geometry	3D geometry	water volume in rolling cylinder
2019	Guts	team-problems	4	Combinatorics	Counting & arrangements	arrangements of letters with repeats
2019	Guts	team-problems	5	Geometry	Coordinate & analytic geometry	lattice points from rotated sublattice, distance
2019	Guts	team-problems	6	Combinatorics	Probability	random pointing teams after many turns
2019	Guts	team-problems	7	Number Theory	Diophantine equations	b^8+m^4+t^2+1=2019 integer triples
2019	Guts	team-problems	8	Number Theory	Diophantine equations	k_i such that quadratic has rational roots
2019	Guts	team-problems	9	Combinatorics	Graphs, grids, colorings & tilings	coloring cube vertices, edges, faces, interior
2019	Guts	team-problems	10	Number Theory	Modular arithmetic	product of units mod 2019, Wilson
2019	Guts	team-problems	11	Combinatorics	Counting & arrangements	bit-pattern baseball matchups highest score
2019	Guts	team-problems	12	Combinatorics	Games, invariants & pigeonhole	paintball duel logical strategy
2019	Guts	team-problems	13	Geometry	Circles (tangents, power of a point, cyclic)	incircle, circumcircle, area of ABPC
2019	Guts	team-problems	14	Geometry	Transformations, loci & advanced tools	laser reflections unfolded in hexagon lattice
2019	Guts	team-problems	15	Other	Logic & puzzles	decode alien operations
2020	Algebra	algebra-problems	1	Algebra	Sequences, series & recurrences	Juggling toss count, arithmetic-type sum
2020	Algebra	algebra-problems	2	Algebra	Polynomials & Vieta	Vieta, a/b+b/a given
2020	Algebra	algebra-problems	3	Algebra	Polynomials & Vieta	Degree 2021 polynomial with 2020 roots, mean of values
2020	Algebra	algebra-problems	4	Algebra	Sequences, series & recurrences	Golden ratio powers, Fibonacci-like coefficients
2020	Algebra	algebra-problems	5	Algebra	Functions & functional equations	Functional equation f(x)f(y)=f(xy)+f(x/y)
2020	Algebra	algebra-problems	6	Algebra	Sequences, series & recurrences	Binomial sum telescoping
2020	Algebra	algebra-problems	7	Algebra	Polynomials & Vieta	Symmetric expressions given a+b+c, abc
2020	Algebra	algebra-problems	8	Algebra	Inequalities & optimization	Smallest constant C in two-variable inequality
2020	Algebra	algebra-problems	9	Number Theory	Diophantine equations	Factor a^3+3b^3+9c^3-9abc identity
2020	Algebra	algebra-problems	10	Algebra	Trigonometry (algebraic)	Telescoping sum of arccos terms
2020	Calculus	calculus-problems	1	Calculus	Derivatives & optimization	Second derivative of monomial
2020	Calculus	calculus-problems	2	Calculus	Integration techniques	Integral of 20-
2020	Calculus	calculus-problems	3	Calculus	Derivatives & optimization	Self-referential f with f'(2)
2020	Calculus	calculus-problems	4	Calculus	Applications (area, volume, rates)	Area between parabola, tangent line, y-axis
2020	Calculus	calculus-problems	5	Combinatorics	Probability	Geometric probability on area of parallelogram, integral
2020	Calculus	calculus-problems	6	Calculus	Derivatives & optimization	Tangent curves y=a^x and log_a x
2020	Calculus	calculus-problems	7	Calculus	Limits	Limit of scaled integral near zero
2020	Calculus	calculus-problems	8	Calculus	Series (calculus)	Double infinite sum
2020	Calculus	calculus-problems	9	Calculus	Differential equations & multivariable	Functional-differential equation for f
2020	Calculus	calculus-problems	10	Calculus	Advanced / tricky integrals	Integral of x/tan(x)

Year	Test	Source file	#	Subject	Topic	Key idea
2020	Discrete	discrete-problems	1	Combinatorics	Counting & arrangements	Permutations with restricted first element
2020	Discrete	discrete-problems	2	Combinatorics	Probability	Two integers, product divisible by 4
2020	Discrete	discrete-problems	3	Number Theory	Modular arithmetic	Wilson's theorem mod 101
2020	Discrete	discrete-problems	4	Combinatorics	Expected value	Expected lights on after three toggles
2020	Discrete	discrete-problems	5	Combinatorics	Probability	Sign of product of random reals
2020	Discrete	discrete-problems	6	Number Theory	Arithmetic functions & valuations	Largest power of 3 dividing subset count
2020	Discrete	discrete-problems	7	Number Theory	Diophantine equations	Count triples satisfying symmetric equation
2020	Discrete	discrete-problems	8	Combinatorics	Graphs, grids, colorings & tilings	Tree of recruits, maximize executives
2020	Discrete	discrete-problems	9	Other	Linear algebra & misc	Linear map on torus, periodic points
2020	Discrete	discrete-problems	10	Number Theory	Modular arithmetic	Count quadratic residue-like values mod n
2020	General	individual-problems	1	Combinatorics	Counting & arrangements	Sequences of three die rolls
2020	General	individual-problems	2	Other	Logic & puzzles	Self-referential answer equation
2020	General	individual-problems	3	Algebra	Arithmetic & word problems	Proportional radius to sqrt of area
2020	General	individual-problems	4	Algebra	Equations & systems	Integer linear system
2020	General	individual-problems	5	Geometry	3D geometry	Cylinder slicing surface area increase
2020	General	individual-problems	6	Combinatorics	Probability	Two integers, product divisible by 4
2020	General	individual-problems	7	Geometry	Coordinate & analytic geometry	Circle intersecting square at six points
2020	General	individual-problems	8	Combinatorics	Counting & arrangements	Hex passcodes with exactly two prime digits
2020	General	individual-problems	9	Geometry	Circles (tangents, power of a point, cyclic)	Circle tangent to circumcircle and triangle side
2020	General	individual-problems	10	Number Theory	Divisibility, primes & factorization	Prime p with p^4+34 prime
2020	General	individual-problems	11	Geometry	Polygons & area	Lune area from equilateral triangle
2020	General	individual-problems	12	Number Theory	Modular arithmetic	Wilson's theorem mod 101
2020	General	individual-problems	13	Geometry	Polygons & area	Fraction of hexagon in strips
2020	General	individual-problems	14	Algebra	Polynomials & Vieta	Vieta on cubic, product of (root+1)
2020	General	individual-problems	15	Algebra	Polynomials & Vieta	Degree 2021 polynomial with 2020 roots
2020	General	individual-problems	16	Geometry	Triangles (lengths, angles, areas)	Triangle with area 6, two third sides
2020	General	individual-problems	17	Combinatorics	Probability	Balls in bins, exactly one empty
2020	General	individual-problems	18	Number Theory	Modular arithmetic	Small modular system, sum of x+y
2020	General	individual-problems	19	Geometry	Circles (tangents, power of a point, cyclic)	Visible area around pillar
2020	General	individual-problems	20	Number Theory	Modular arithmetic	CRT count of odd remainders
2020	General	individual-problems	21	Combinatorics	Probability	Sign of product of random reals
2020	General	individual-problems	22	Combinatorics	Expected value	Expected lights on after three toggles
2020	General	individual-problems	23	Geometry	Circles (tangents, power of a point, cyclic)	Circumcenter of triangle in circle, ratio
2020	General	individual-problems	24	Number Theory	Arithmetic functions & valuations	Largest power of 3 dividing subset count
2020	General	individual-problems	25	Algebra	Functions & functional equations	Functional equation f(x)f(y)=f(xy)+f(x/y)
2020	Geometry	geometry-problems	1	Geometry	3D geometry	Cylinder slicing surface area increase
2020	Geometry	geometry-problems	2	Geometry	Circles (tangents, power of a point, cyclic)	Tangent circles on diameter, minimize area sum
2020	Geometry	geometry-problems	3	Geometry	3D geometry	Right prism volume with angles

Year	Test	Source file	#	Subject	Topic	Key idea
2020	Geometry	geometry-problems	4	Geometry	Circles (tangents, power of a point, cyclic)	Visible area around pillar, tangent lines
2020	Geometry	geometry-problems	5	Geometry	Polygons & area	Nested squares, geometric series of areas
2020	Geometry	geometry-problems	6	Geometry	3D geometry	Isosceles tetrahedron volume
2020	Geometry	geometry-problems	7	Geometry	Circles (tangents, power of a point, cyclic)	Circumcenter of triangle in circle, ratio
2020	Geometry	geometry-problems	8	Geometry	Similarity, ratios & mass points	Isogonal cevians concurrency, area ratios
2020	Geometry	geometry-problems	9	Geometry	Polygons & area	Minimum area convex polygon with Fibonacci sides
2020	Geometry	geometry-problems	10	Geometry	Coordinate & analytic geometry	Ellipse foci, perimeter of quadrilateral
2020	Guts	team-problems	1	Combinatorics	Probability	Same number chosen from 1 to 10
2020	Guts	team-problems	2	Algebra	Arithmetic & word problems	Teams of 9, even number of teams
2020	Guts	team-problems	3	Geometry	3D geometry	Shortest path on cube surface
2020	Guts	team-problems	4	Number Theory	Divisibility, primes & factorization	Largest prime divisor of difference
2020	Guts	team-problems	5	Number Theory	Divisibility, primes & factorization	Sums of squares of distinct primes
2020	Guts	team-problems	6	Number Theory	Digits & bases	Numbers avoiding digits 2 and 9
2020	Guts	team-problems	7	Combinatorics	Probability	Increasing sequence of die rolls
2020	Guts	team-problems	8	Geometry	Polygons & area	Square split into regions with equal areas
2020	Guts	team-problems	9	Algebra	Sequences, series & recurrences	Alternating sum of squares recurrence
2020	Guts	team-problems	10	Combinatorics	Counting & arrangements	Three-digit numbers with exactly two repeated digits
2020	Guts	team-problems	11	Number Theory	Divisibility, primes & factorization	Sum of gcd(x,10x+9)
2020	Guts	team-problems	12	Algebra	Inequalities & optimization	Minimize surface area of open box, fixed volume
2020	Guts	team-problems	13	Combinatorics	Expected value	Expected sum of random subset
2020	Guts	team-problems	14	Geometry	Polygons & area	Star-shaped figure area
2020	Guts	team-problems	15	Combinatorics	Expected value	Expected self-locating substrings
2020	Guts	team-problems	16	Geometry	Coordinate & analytic geometry	Rectangle with equal segments, right angle
2020	Guts	team-problems	17	Algebra	Polynomials & Vieta	Real roots of palindromic quartic
2020	Guts	team-problems	18	Combinatorics	Expected value	Expected count of two-same triples in flips
2020	Guts	team-problems	19	Combinatorics	Expected value	Expected flips with decreasing success chance
2020	Guts	team-problems	20	Geometry	Triangles (lengths, angles, areas)	Kite-like quadrilateral integer sides
2020	Guts	team-problems	21	Geometry	Circles (tangents, power of a point, cyclic)	AI times IX via power of point
2020	Guts	team-problems	22	Geometry	Triangles (lengths, angles, areas)	System as law of cosines, 120 degree triangle
2020	Guts	team-problems	23	Algebra	Trigonometry (algebraic)	Sum of cosines vanishing, find theta
2020	Guts	team-problems	24	Algebra	Sequences, series & recurrences	Binomial sum, smallest N inequality
2020	Guts	team-problems	25	Other	Estimation	Unique-submission game
2020	Guts	team-problems	26	Other	Estimation	Estimate 2020th twin prime
2020	Guts	team-problems	27	Other	Estimation	Count of 1s in hex of 2020!
2021	Algebra	algebra-problems	1	Algebra	Equations & systems	Two quadratic equations in x, compute x^4
2021	Algebra	algebra-problems	2	Algebra	Functions & functional equations	Linear functions commutator difference
2021	Algebra	algebra-problems	3	Algebra	Floor, ceiling & absolute value	Nested floor equation, upper bound of solutions
2021	Algebra	algebra-problems	4	Algebra	Trigonometry (algebraic)	Sin 2θ equation, maximize (1+sin)(1+cos)

Year	Test	Source file	#	Subject	Topic	Key idea
2021	Algebra	algebra-problems	5	Algebra	Floor, ceiling & absolute value	Floor times fractional part equation
2021	Algebra	algebra-problems	6	Algebra	Functions & functional equations	Functional equation with transformations of x
2021	Algebra	algebra-problems	7	Algebra	Complex numbers & roots of unity	Roots of unity sum of 1/(1-z^2020)
2021	Algebra	algebra-problems	8	Algebra	Polynomials & Vieta	Cubic roots as right triangle sides
2021	Algebra	algebra-problems	9	Algebra	Polynomials & Vieta	Reducibility of x^n+x+1
2021	Algebra	algebra-problems	10	Number Theory	Arithmetic functions & valuations	Square-free counting sums, sign of difference
2021	Calculus	calculus-problems	1	Calculus	Integration techniques	Fundamental theorem of calculus derivative
2021	Calculus	calculus-problems	2	Calculus	Derivatives & optimization	Fourth derivative of product of linears
2021	Calculus	calculus-problems	3	Calculus	Derivatives & optimization	Quadratic and derivative intersect once
2021	Calculus	calculus-problems	4	Calculus	Applications (area, volume, rates)	Area of region bounded by parabolas
2021	Calculus	calculus-problems	5	Calculus	Integration techniques	Trig integral antiderivative matching
2021	Calculus	calculus-problems	6	Calculus	Limits	Iterated sin(arccos) limit
2021	Calculus	calculus-problems	7	Calculus	Integration techniques	Integral of inverse function
2021	Calculus	calculus-problems	8	Calculus	Integration techniques	Integral of t^(6n-1)/(1+t^3), limit
2021	Calculus	calculus-problems	9	Calculus	Applications (area, volume, rates)	Supremum of accumulated score, integral-like
2021	Calculus	calculus-problems	10	Calculus	Series (calculus)	Infinite product with arctan
2021	Discrete	discrete-problems	1	Combinatorics	Counting & arrangements	Three-card hands with both colors
2021	Discrete	discrete-problems	2	Number Theory	Modular arithmetic	Two arithmetic progressions common terms
2021	Discrete	discrete-problems	3	Combinatorics	Counting & arrangements	Distinct sums from 8 chosen numbers
2021	Discrete	discrete-problems	4	Combinatorics	Probability	Two players both captains
2021	Discrete	discrete-problems	5	Number Theory	Digits & bases	Palindromic sum of number and reversal
2021	Discrete	discrete-problems	6	Number Theory	Arithmetic functions & valuations	Number of divisors of n^n
2021	Discrete	discrete-problems	7	Combinatorics	Recursion & states	Minimum operations to reach 1
2021	Discrete	discrete-problems	8	Combinatorics	Expected value	Expected value of random nondecreasing decimal
2021	Discrete	discrete-problems	9	Number Theory	Modular arithmetic	Harmonic sum mod prime 101
2021	Discrete	discrete-problems	10	Combinatorics	Graphs, grids, colorings & tilings	Triangle-free graph with 22 edges, Turan
2021	General	general-problems	1	Algebra	Arithmetic & word problems	Ratio of school sizes
2021	General	general-problems	2	Algebra	Equations & systems	Two quadratic equations in x, compute x^4
2021	General	general-problems	3	Geometry	Triangles (lengths, angles, areas)	Smallest angle in scalene acute integer triangle
2021	General	general-problems	4	Algebra	Arithmetic & word problems	Work rate problem with tea drinking
2021	General	general-problems	5	Combinatorics	Counting & arrangements	Distinct pairs of boards from one cut
2021	General	general-problems	6	Geometry	3D geometry	Surface area of toilet paper roll
2021	General	general-problems	7	Number Theory	Modular arithmetic	Two arithmetic progressions common terms
2021	General	general-problems	8	Algebra	Arithmetic & word problems	Pie chart angle change
2021	General	general-problems	9	Combinatorics	Probability	HHH before THH
2021	General	general-problems	10	Geometry	Similarity, ratios & mass points	Similar triangles in isosceles triangle
2021	General	general-problems	11	Combinatorics	Inclusion-exclusion, bijections & lattice paths	Sequences with gcd 1, inclusion-exclusion
2021	General	general-problems	12	Algebra	Polynomials & Vieta	Sum of reciprocals of cubic roots

Year	Test	Source file	#	Subject	Topic	Key idea
2021	General	general-problems	13	Combinatorics	Probability	Minimum of four die rolls is 4
2021	General	general-problems	14	Algebra	Complex numbers & roots of unity	Sum of powers of 47th roots of unity
2021	General	general-problems	15	Combinatorics	Expected value	Chip-firing expected number of chips
2021	General	general-problems	16	Combinatorics	Probability	Meeting time with unequal waiting, geometric probability
2021	General	general-problems	17	Algebra	Arithmetic & word problems	Simplify nested radicals difference
2021	General	general-problems	18	Geometry	Transformations, loci & advanced tools	Reflection for equal angles, AE+EB
2021	General	general-problems	19	Number Theory	Digits & bases	Palindromic sum of number and reversal
2021	General	general-problems	20	Algebra	Complex numbers & roots of unity	Geometric series of (1+i)^k
2021	General	general-problems	21	Number Theory	Modular arithmetic	Binomial power of 1+sqrt2 mod 13
2021	General	general-problems	22	Combinatorics	Probability	Random flights with distinct cities
2021	General	general-problems	23	Combinatorics	Counting & arrangements	Square symmetry sequences returning to identity
2021	General	general-problems	24	Number Theory	Diophantine equations	xyz=2(x+y+z) integer solutions
2021	General	general-problems	25	Geometry	Similarity, ratios & mass points	Nested centroids, area limit pattern
2021	Geometry	geometry-problems	1	Geometry	Polygons & area	Rectangle cut maximizing perimeter
2021	Geometry	geometry-problems	2	Geometry	Triangles (lengths, angles, areas)	Smallest triangle containing six unit circles
2021	Geometry	geometry-problems	3	Geometry	Transformations, loci & advanced tools	Reflection for equal angles, AE+EB
2021	Geometry	geometry-problems	4	Geometry	Polygons & area	Equilateral polygon with 15 and 300 degree angles
2021	Geometry	geometry-problems	5	Geometry	Circles (tangents, power of a point, cyclic)	Two circles, common tangent, power of point
2021	Geometry	geometry-problems	6	Geometry	3D geometry	Shortest ant path on modified cube
2021	Geometry	geometry-problems	7	Geometry	Polygons & area	Regular hexagon area from distances to line
2021	Geometry	geometry-problems	8	Geometry	Circles (tangents, power of a point, cyclic)	Circumcircles through O, line perpendicular to BC
2021	Geometry	geometry-problems	9	Geometry	Transformations, loci & advanced tools	Equilateral triangles, rotation, area of midpoint triangle
2021	Geometry	geometry-problems	10	Geometry	Transformations, loci & advanced tools	Minimize sum of distances via reflections, incircle
2021	Guts	guts-problems	1	Algebra	Arithmetic & word problems	Linear interpolation of charge versus pH
2021	Guts	guts-problems	2	Algebra	Arithmetic & word problems	Battery time with stepwise brightness
2021	Guts	guts-problems	3	Algebra	Logarithms & exponents	Log product simplification
2021	Guts	guts-problems	4	Algebra	Logarithms & exponents	Exponential equation as quadratic in 2^x
2021	Guts	guts-problems	5	Geometry	3D geometry	Shortest path on tetrahedron avoiding a face
2021	Guts	guts-problems	6	Combinatorics	Probability	Three numbers with integer mean
2021	Guts	guts-problems	7	Number Theory	Divisibility, primes & factorization	Weekly increase by largest prime factor
2021	Guts	guts-problems	8	Geometry	Circles (tangents, power of a point, cyclic)	Tangent circles, common external tangent triangle
2021	Guts	guts-problems	9	Combinatorics	Graphs, grids, colorings & tilings	Minimum boundary in three-coloring of grid
2021	Guts	guts-problems	10	Combinatorics	Counting & arrangements	Subsets where mean of max and min is in set
2021	Guts	guts-problems	11	Number Theory	Modular arithmetic	Primes dividing sum of powers
2021	Guts	guts-problems	12	Geometry	Similarity, ratios & mass points	Square folded construction, similar triangles
2021	Guts	guts-problems	13	Combinatorics	Graphs, grids, colorings & tilings	Colorings of 3x3 grid with no equal adjacent
2021	Guts	guts-problems	14	Number Theory	Modular arithmetic	Iterated recurrence mod 47, orders
2021	Guts	guts-problems	15	Algebra	Trigonometry (algebraic)	Infinite product of cosines ratio

Year	Test	Source file	#	Subject	Topic	Key idea
2021	Guts	guts-problems	16	Combinatorics	Expected value	Expected time with fatigue probability
2021	Guts	guts-problems	17	Geometry	Triangles (lengths, angles, areas)	Orthocenter, circumcenter, OH parallel to BC
2021	Guts	guts-problems	18	Algebra	Equations & systems	Nested cube root equation
2021	Guts	guts-problems	19	Algebra	Equations & systems	Radical equation in a, with chained answers
2021	Guts	guts-problems	20	Geometry	Triangles (lengths, angles, areas)	Incircle and excircle, area of quadrilateral
2021	Guts	guts-problems	21	Geometry	Coordinate & analytic geometry	Lines of slopes b/c, intersection count in unit square
2021	Guts	guts-problems	22	Geometry	Triangles (lengths, angles, areas)	Angle bisector, perpendiculars, distance to line
2021	Guts	guts-problems	23	Combinatorics	Inclusion-exclusion, bijections & lattice paths	Lattice paths avoiding a modular line, generating sum
2021	Guts	guts-problems	24	Number Theory	Diophantine equations	a^2+b^2+c^2-ab-bc-ca=2021p
2021	Guts	guts-problems	25	Other	Estimation	Period of decimal expansion of 2021^2021/2021!
2021	Guts	guts-problems	26	Other	Estimation	Expected value of random walk to square
2021	Guts	guts-problems	27	Other	Estimation	Digit parity count across powers of 2
2022	Algebra	algebra-problems	1	Algebra	Sequences, series & recurrences	Iterated linear operation
2022	Algebra	algebra-problems	2	Algebra	Logarithms & exponents	Exponential equation as quadratic in 2^x
2022	Algebra	algebra-problems	3	Algebra	Sequences, series & recurrences	Geometric then arithmetic sequence, minimize d/a
2022	Algebra	algebra-problems	4	Algebra	Floor, ceiling & absolute value	Floor with ceiling nested equation, intervals
2022	Algebra	algebra-problems	5	Algebra	Inequalities & optimization	Extremes of T given sum and sum of squares
2022	Algebra	algebra-problems	6	Algebra	Polynomials & Vieta	Degree-6 polynomial from difference values
2022	Algebra	algebra-problems	7	Algebra	Polynomials & Vieta	Vieta on cubic, sum of 1/(1-r^2)
2022	Algebra	algebra-problems	8	Algebra	Polynomials & Vieta	Products of shifted variables as polynomial values
2022	Algebra	algebra-problems	9	Algebra	Trigonometry (algebraic)	Tangent addition recurrence, first undefined term
2022	Algebra	algebra-problems	10	Algebra	Polynomials & Vieta	Vieta cyclic expressions on cubic
2022	Calculus	calculus-problems	1	Calculus	Derivatives & optimization	Derivative of sin^2 plus cos^2
2022	Calculus	calculus-problems	2	Calculus	Derivatives & optimization	Equal derivative values of cubic
2022	Calculus	calculus-problems	3	Calculus	Integration techniques	Substitution with nested exponentials
2022	Calculus	calculus-problems	4	Calculus	Derivatives & optimization	Polynomial with shared roots of f and f'
2022	Calculus	calculus-problems	5	Calculus	Series (calculus)	Telescoping infinite sum of radicals
2022	Calculus	calculus-problems	6	Calculus	Integration techniques	Substitution in trig integral
2022	Calculus	calculus-problems	7	Calculus	Limits	Limit of 1+integral to power 1/x
2022	Calculus	calculus-problems	8	Combinatorics	Expected value	Expected number of uniform purchases exceeding 1
2022	Calculus	calculus-problems	9	Calculus	Advanced / tricky integrals	Integral of cot(x)ln(cos x)
2022	Calculus	calculus-problems	10	Calculus	Applications (area, volume, rates)	Volume swept by rotating cube
2022	Discrete	discrete-problems	1	Combinatorics	Counting & arrangements	Lunch choices with constraints
2022	Discrete	discrete-problems	2	Number Theory	Divisibility, primes & factorization	Divisors of 100000 without digit 0
2022	Discrete	discrete-problems	3	Combinatorics	Probability	Dice sum versus coin product
2022	Discrete	discrete-problems	4	Combinatorics	Counting & arrangements	Arm wrestling outcomes with more than 3 streaks
2022	Discrete	discrete-problems	5	Number Theory	Digits & bases	Digit sum equation for n
2022	Discrete	discrete-problems	6	Combinatorics	Probability	Quadratic with distinct real roots from powers of 2

Year	Test	Source file	#	Subject	Topic	Key idea
2022	Discrete	discrete-problems	7	Combinatorics	Expected value	Expected cards flipped until ace and king
2022	Discrete	discrete-problems	8	Number Theory	Modular arithmetic	Linear recurrence pair mod 31
2022	Discrete	discrete-problems	9	Combinatorics	Inclusion-exclusion, bijections & lattice paths	Selecting even-sized subsets, counting mod 42^2-1
2022	Discrete	discrete-problems	10	Number Theory	Diophantine equations	Pairs with a^2+b^2 dividing 10!
2022	General	general-problems	1	Number Theory	Digits & bases	Two-digit odd numbers with large digits
2022	General	general-problems	2	Algebra	Arithmetic & word problems	Iterated linear operation
2022	General	general-problems	3	Combinatorics	Counting & arrangements	Lunch choices with constraints
2022	General	general-problems	4	Geometry	Polygons & area	Shaded area in 4x4 grid
2022	General	general-problems	5	Number Theory	Modular arithmetic	Last digit of power expression
2022	General	general-problems	6	Geometry	Polygons & area	Paper folding, longest distance between corners
2022	General	general-problems	7	Algebra	Floor, ceiling & absolute value	Fractional part squared via floor
2022	General	general-problems	8	Combinatorics	Probability	Dice game ending with $3
2022	General	general-problems	9	Algebra	Arithmetic & word problems	Clock hands angle over time interval
2022	General	general-problems	10	Algebra	Arithmetic & word problems	Fill equation with distinct numbers
2022	General	general-problems	11	Algebra	Logarithms & exponents	Exponential equation as quadratic in 2^x
2022	General	general-problems	12	Geometry	Polygons & area	Parallelogram angles in hexagon
2022	General	general-problems	13	Combinatorics	Probability	Product of three dice is a cube
2022	General	general-problems	14	Number Theory	Divisibility, primes & factorization	Divisors of 100000 without digit 0
2022	General	general-problems	15	Geometry	Polygons & area	Tilted square circumscribing a square
2022	General	general-problems	16	Algebra	Inequalities & optimization	Median minimizes total walking distance
2022	General	general-problems	17	Combinatorics	Counting & arrangements	Assigning chores with constraints
2022	General	general-problems	18	Geometry	Circles (tangents, power of a point, cyclic)	Tangent circles in circumcircle, shaded area
2022	General	general-problems	19	Algebra	Sequences, series & recurrences	Geometric then arithmetic sequence, minimize d/a
2022	General	general-problems	20	Number Theory	Modular arithmetic	Differences covering all residues mod 57
2022	General	general-problems	21	Geometry	Polygons & area	Area swept by moving triangle in hexagon
2022	General	general-problems	22	Number Theory	Digits & bases	Digit sum equation for n
2022	General	general-problems	23	Algebra	Inequalities & optimization	Extremes of T given sum and sum of squares
2022	General	general-problems	24	Geometry	Triangles (lengths, angles, areas)	Incenter on line parallel to side, find AY
2022	General	general-problems	25	Combinatorics	Probability	Quadratic with distinct real roots from powers of 2
2022	Geometry	geometry-problems	1	Geometry	Polygons & area	Paper folding, longest distance between corners
2022	Geometry	geometry-problems	2	Geometry	Polygons & area	Tilted square circumscribing a square
2022	Geometry	geometry-problems	3	Geometry	Circles (tangents, power of a point, cyclic)	Tangent circles in circumcircle, shaded area
2022	Geometry	geometry-problems	4	Geometry	Polygons & area	Area swept by moving triangle in hexagon
2022	Geometry	geometry-problems	5	Geometry	3D geometry	Tilted tricycle wheels tracing circles
2022	Geometry	geometry-problems	6	Geometry	Triangles (lengths, angles, areas)	Incenter on line parallel to side, find AY
2022	Geometry	geometry-problems	7	Geometry	Triangles (lengths, angles, areas)	Orthocenter, angle bisector, find BC
2022	Geometry	geometry-problems	8	Geometry	Polygons & area	Nested polygons and circles, limiting area
2022	Geometry	geometry-problems	9	Geometry	Transformations, loci & advanced tools	Radical planes of seven spheres

Year	Test	Source file	#	Subject	Topic	Key idea
2022	Geometry	geometry-problems	10	Geometry	Circles (tangents, power of a point, cyclic)	Tangent at A, orthic triangle line, find XY
2022	Guts	guts-problems	1	Number Theory	Digits & bases	Two-digit numbers with digit sum 16
2022	Guts	guts-problems	2	Combinatorics	Probability	Marbles drawn without replacement
2022	Guts	guts-problems	3	Algebra	Polynomials & Vieta	Quadratic from roots and y-intercept
2022	Guts	guts-problems	4	Number Theory	Divisibility, primes & factorization	Primes below 40 that are sum of two primes
2022	Guts	guts-problems	5	Algebra	Arithmetic & word problems	Commute rates and times
2022	Guts	guts-problems	6	Geometry	Polygons & area	Square and equilateral triangle, area
2022	Guts	guts-problems	7	Geometry	Polygons & area	Inner hexagon formed by diagonals
2022	Guts	guts-problems	8	Combinatorics	Probability	Triangle from heptagon vertices contains center
2022	Guts	guts-problems	9	Number Theory	Modular arithmetic	Polynomial values mod 5 over range
2022	Guts	guts-problems	10	Number Theory	Arithmetic functions & valuations	Coprime divisor pairs of 5040
2022	Guts	guts-problems	11	Number Theory	Digits & bases	Count digit 2 in multiples of 125
2022	Guts	guts-problems	12	Geometry	Circles (tangents, power of a point, cyclic)	Internally tangent circles, perpendicular and tangent line
2022	Guts	guts-problems	13	Algebra	Equations & systems	Symmetric cubic system in x and y
2022	Guts	guts-problems	14	Number Theory	Arithmetic functions & valuations	Sum of reduced denominators, gcd with 300
2022	Guts	guts-problems	15	Combinatorics	Counting & arrangements	Maximize substrings containing a 1
2022	Guts	guts-problems	16	Geometry	Triangles (lengths, angles, areas)	Incenter, circles centered at vertices, triangle area
2022	Guts	guts-problems	17	Algebra	Equations & systems	Cyclic cubic system, count integer triples
2022	Guts	guts-problems	18	Algebra	Complex numbers & roots of unity	Roots of unity filter on binomial expansion
2022	Guts	guts-problems	19	Algebra	Complex numbers & roots of unity	Sum of squared chord products in regular polygon
2022	Guts	guts-problems	20	Algebra	Complex numbers & roots of unity	Expected product over roots of unity
2022	Guts	guts-problems	21	Number Theory	Modular arithmetic	Primes dividing values of cyclotomic polynomial
2022	Guts	guts-problems	22	Number Theory	Modular arithmetic	Sum over divisors with binomial mod 2n
2022	Guts	guts-problems	23	Combinatorics	Expected value	Expected components after row/column harvest
2022	Guts	guts-problems	24	Geometry	Transformations, loci & advanced tools	Equilateral circumscribed triangle, projective construction, altitudes
2022	Guts	guts-problems	25	Other	Estimation	Veryexcircle triangle area threshold
2022	Guts	guts-problems	26	Other	Estimation	Count n with digits of floor expression
2022	Guts	guts-problems	27	Other	Estimation	Submit number, sum of squares of digits game
2023	Algebra	algebra-problems	1	Algebra	Equations & systems	Two lines meeting at a point, find slopes
2023	Algebra	algebra-problems	2	Algebra	Floor, ceiling & absolute value	Cases on |x|,|y| signs, |x-y|
2023	Algebra	algebra-problems	3	Algebra	Sequences, series & recurrences	Geometric sequences with equal ratio
2023	Algebra	algebra-problems	4	Algebra	Floor, ceiling & absolute value	Piecewise linear f via floor, evaluate polynomial
2023	Algebra	algebra-problems	5	Algebra	Polynomials & Vieta	Vieta on shifted roots
2023	Algebra	algebra-problems	6	Algebra	Sequences, series & recurrences	Recurrence generating function sum equals 1
2023	Algebra	algebra-problems	7	Algebra	Polynomials & Vieta	Count polynomial pairs a^2+b^2 factorization
2023	Algebra	algebra-problems	8	Algebra	Polynomials & Vieta	Cubic roots as right triangle sides, minimize t
2023	Algebra	algebra-problems	9	Algebra	Functions & functional equations	Iterating logistic map, count preimages
2023	Algebra	algebra-problems	10	Algebra	Equations & systems	Cubic form factors, integer solution triple

Year	Test	Source file	#	Subject	Topic	Key idea
2023	Calculus	calculus-problems	1	Calculus	Derivatives & optimization	Derivative of trig at pi/4
2023	Calculus	calculus-problems	2	Calculus	Integration techniques	Integrate polynomial in (x-2), odd symmetry
2023	Calculus	calculus-problems	3	Calculus	Integration techniques	Area under sqrt(kx)
2023	Calculus	calculus-problems	4	Calculus	Applications (area, volume, rates)	Related rates, melting cube
2023	Calculus	calculus-problems	5	Algebra	Inequalities & optimization	Riemann sum lower bound on cubic form
2023	Calculus	calculus-problems	6	Calculus	Derivatives & optimization	nth derivatives of 1/x, sum series
2023	Calculus	calculus-problems	7	Calculus	Series (calculus)	Series sum n a_n/6^n via recurrence
2023	Calculus	calculus-problems	8	Algebra	Inequalities & optimization	Maximize integral pointwise over f, completing square
2023	Calculus	calculus-problems	9	Calculus	Integration techniques	Integral of (sin+cos)^6 over period
2023	Calculus	calculus-problems	10	Calculus	Advanced / tricky integrals	Integral of sin x/x^2 with series of sin(nx)/n!
2023	Discrete	discrete-problems	1	Number Theory	Digits & bases	Three-digit number divisibility and remainder
2023	Discrete	discrete-problems	2	Combinatorics	Counting & arrangements	Enter/exit orderings of names
2023	Discrete	discrete-problems	3	Number Theory	Digits & bases	Count n where digit 4s increase on adding one
2023	Discrete	discrete-problems	4	Number Theory	Modular arithmetic	Nested remainders mod 11,5,2 count
2023	Discrete	discrete-problems	5	Combinatorics	Probability	Geometric N then N rolls, probability of 6
2023	Discrete	discrete-problems	6	Number Theory	Arithmetic functions & valuations	gcd lcm condition counting via CRT
2023	Discrete	discrete-problems	7	Combinatorics	Expected value	Expected extra area, random chord of square
2023	Discrete	discrete-problems	8	Number Theory	Arithmetic functions & valuations	Divisor-sum function iterates mod 1001
2023	Discrete	discrete-problems	9	Combinatorics	Counting & arrangements	Counting permutations with three monotone sections
2023	Discrete	discrete-problems	10	Number Theory	Modular arithmetic	Power of quadratic irrational mod prime
2023	General	general-problems	1	Algebra	Arithmetic & word problems	Chocolates ratio word problem
2023	General	general-problems	2	Algebra	Arithmetic & word problems	Telescoping alternating products sum
2023	General	general-problems	3	Geometry	Triangles (lengths, angles, areas)	Semicircle in rectangle, diagonal
2023	General	general-problems	4	Algebra	Equations & systems	Linear system, sum of variables
2023	General	general-problems	5	Algebra	Equations & systems	Two lines intersecting, slopes sum
2023	General	general-problems	6	Number Theory	Digits & bases	Digit-constrained three-digit number
2023	General	general-problems	7	Algebra	Sequences, series & recurrences	Arithmetic sequence middle term sum
2023	General	general-problems	8	Number Theory	Divisibility, primes & factorization	Primes of form n^2+1 below 100
2023	General	general-problems	9	Algebra	Arithmetic & word problems	Maximize/minimize expression with digit assignment
2023	General	general-problems	10	Geometry	Circles (tangents, power of a point, cyclic)	Circle through six trisection points of triangle
2023	General	general-problems	11	Algebra	Sequences, series & recurrences	Geometric sequences with equal ratio
2023	General	general-problems	12	Algebra	Floor, ceiling & absolute value	Greatest integer below sum with radicals
2023	General	general-problems	13	Combinatorics	Counting & arrangements	Enter/exit orderings of names
2023	General	general-problems	14	Algebra	Floor, ceiling & absolute value	Cases on |x|,|y|, |x-y|
2023	General	general-problems	15	Number Theory	Digits & bases	Count n where digit 4s increase
2023	General	general-problems	16	Number Theory	Divisibility, primes & factorization	Smallest n with perfect power conditions, divisor count
2023	General	general-problems	17	Geometry	3D geometry	Frustum plus cylinder volume
2023	General	general-problems	18	Combinatorics	Probability	Geometric N then N rolls, probability of 6

Year	Test	Source file	#	Subject	Topic	Key idea
2023	General	general-problems	19	Geometry	Circles (tangents, power of a point, cyclic)	Tangent lengths, perimeter of triangle around circle
2023	General	general-problems	20	Number Theory	Modular arithmetic	Sum of divisors with digit condition mod 131
2023	General	general-problems	21	Algebra	Polynomials & Vieta	Vieta on shifted roots
2023	General	general-problems	22	Geometry	Circles (tangents, power of a point, cyclic)	Two tangent circles in right triangle, inradius-like
2023	General	general-problems	23	Number Theory	Arithmetic functions & valuations	gcd lcm condition counting
2023	General	general-problems	24	Algebra	Sequences, series & recurrences	Recurrence series sum equals 1
2023	General	general-problems	25	Geometry	3D geometry	Tetrahedron midsphere radius
2023	Geometry	geometry-problems	1	Geometry	Triangles (lengths, angles, areas)	Semicircle in rectangle, diagonal
2023	Geometry	geometry-problems	2	Geometry	Circles (tangents, power of a point, cyclic)	Circle through six trisection points
2023	Geometry	geometry-problems	3	Geometry	3D geometry	Frustum plus cylinder volume
2023	Geometry	geometry-problems	4	Geometry	Circles (tangents, power of a point, cyclic)	Tangent lengths, perimeter of triangle around circle
2023	Geometry	geometry-problems	5	Geometry	Circles (tangents, power of a point, cyclic)	Two tangent circles in right triangle
2023	Geometry	geometry-problems	6	Geometry	Circles (tangents, power of a point, cyclic)	Median extended to circumcircle, sine of central angle
2023	Geometry	geometry-problems	7	Geometry	3D geometry	Tetrahedron midsphere radius
2023	Geometry	geometry-problems	8	Geometry	Triangles (lengths, angles, areas)	Circle meets equilateral triangle, area difference
2023	Geometry	geometry-problems	9	Geometry	Transformations, loci & advanced tools	Isogonal-like point T, midpoint, find BC
2023	Geometry	geometry-problems	10	Geometry	Circles (tangents, power of a point, cyclic)	Circumcenter, circle BOC, perpendiculars, AF*EG
2023	Guts	guts-problems	1	Geometry	Polygons & area	Square, triangle area equal, shaded area
2023	Guts	guts-problems	2	Combinatorics	Counting & arrangements	Palindromic colored tower, adjacent colors differ
2023	Guts	guts-problems	3	Number Theory	Arithmetic functions & valuations	Second smallest number with six divisors
2023	Guts	guts-problems	4	Combinatorics	Probability	Grasshopper random walk, return within 3 hops
2023	Guts	guts-problems	5	Geometry	Coordinate & analytic geometry	Two parabolas symmetric, y-intercepts
2023	Guts	guts-problems	6	Geometry	Circles (tangents, power of a point, cyclic)	Rectangle in circle, midpoint ray, length AX
2023	Guts	guts-problems	7	Number Theory	Digits & bases	Digit product equation n - p(n)=52
2023	Guts	guts-problems	8	Geometry	Circles (tangents, power of a point, cyclic)	Common tangent, circles through centers, distance
2023	Guts	guts-problems	9	Geometry	Coordinate & analytic geometry	Count intercepts of x^a+y^b=1, double sum
2023	Guts	guts-problems	10	Geometry	Triangles (lengths, angles, areas)	Centroid with perpendicular medians, sum squares
2023	Guts	guts-problems	11	Algebra	Equations & systems	x+n/x style equation, find valid n
2023	Guts	guts-problems	12	Combinatorics	Counting & arrangements	Count mountainous 5-digit numbers
2023	Guts	guts-problems	13	Algebra	Inequalities & optimization	Cubic roots as triangle sides, bound ab
2023	Guts	guts-problems	14	Geometry	Circles (tangents, power of a point, cyclic)	Right triangle with circles tangent to circumcircle, area DEF
2023	Guts	guts-problems	15	Number Theory	Digits & bases	Digit sum condition on 7n+1, find a_2023
2023	Guts	guts-problems	16	Combinatorics	Probability	Die then cards, score above median
2023	Guts	guts-problems	17	Number Theory	Digits & bases	Smallest multiple of repunit-like with digits 4,8
2023	Guts	guts-problems	18	Algebra	Sequences, series & recurrences	Recurrence with tangent-like structure, b_2023
2023	Guts	guts-problems	19	Combinatorics	Expected value	Expected balls remaining until monochrome
2023	Guts	guts-problems	20	Algebra	Sequences, series & recurrences	Quadratic sums f_k roots, difference grows
2023	Guts	guts-problems	21	Geometry	Triangles (lengths, angles, areas)	Trapezoid bisectors, incircle tangency quadrilateral area

Year	Test	Source file	#	Subject	Topic	Key idea
2023	Guts	guts-problems	22	Other	Estimation	Sum of decimal digits of primes expansions
2023	Guts	guts-problems	23	Combinatorics	Expected value	Robot random walk expected final position
2023	Guts	guts-problems	24	Algebra	Polynomials & Vieta	Monic polynomial factored into quadratics, sum coefficients
2023	Guts	guts-problems	25	Other	Estimation	Squares packed in right triangle, log estimate
2023	Guts	guts-problems	26	Other	Estimation	Integer-root quadratics ratio, minimum N
2023	Guts	guts-problems	27	Other	Estimation	Count radii with integer isosceles inscribed triangle
2024	Algebra	algebra-problems	1	Algebra	Arithmetic & word problems	Apples word problem, linear equations
2024	Algebra	algebra-problems	2	Algebra	Equations & systems	Rational equations with equal values, substitution
2024	Algebra	algebra-problems	3	Algebra	Sequences, series & recurrences	Arithmetic plus geometric sequences
2024	Algebra	algebra-problems	4	Algebra	Floor, ceiling & absolute value	Floor and ceiling equation, greatest x
2024	Algebra	algebra-problems	5	Algebra	Polynomials & Vieta	Cubic roots equidistant from 2024, unique pair
2024	Algebra	algebra-problems	6	Algebra	Equations & systems	Symmetric three-variable quadratic system
2024	Algebra	algebra-problems	7	Algebra	Equations & systems	Triples with symmetric equation, AM-GM equality
2024	Algebra	algebra-problems	8	Algebra	Inequalities & optimization	Min of linear form under symmetric constraints
2024	Algebra	algebra-problems	9	Algebra	Sequences, series & recurrences	Trig-substitution recursions, bounded differences
2024	Algebra	algebra-problems	10	Algebra	Complex numbers & roots of unity	Product over fifth roots of unity with sec, mod 101
2024	Calculus	calculus-problems	1	Calculus	Limits	Limit of cubic difference quotient
2024	Calculus	calculus-problems	2	Calculus	Derivatives & optimization	Derivative of inverse-like composed function
2024	Calculus	calculus-problems	3	Calculus	Derivatives & optimization	Third derivative of rational function
2024	Calculus	calculus-problems	4	Calculus	Derivatives & optimization	Tangency of exponential and root, unique solution
2024	Calculus	calculus-problems	5	Calculus	Integration techniques	Integral of x^floor(x) asymptotics
2024	Calculus	calculus-problems	6	Calculus	Integration techniques	Integral with nested radicals, rational result
2024	Calculus	calculus-problems	7	Calculus	Integration techniques	Integral by substitution with half powers
2024	Calculus	calculus-problems	8	Calculus	Series (calculus)	Alternating series telescoping to logs
2024	Calculus	calculus-problems	9	Calculus	Derivatives & optimization	Newton iteration basin convergence
2024	Calculus	calculus-problems	10	Calculus	Limits	Recursive f with ln 2, average error limit
2024	Discrete	discrete-problems	1	Number Theory	Digits & bases	Third-largest palindromic multiple of three
2024	Discrete	discrete-problems	2	Combinatorics	Counting & arrangements	Periodic coloring with color usage limits
2024	Discrete	discrete-problems	3	Number Theory	Divisibility, primes & factorization	Divisors of 44 with 24-n
2024	Discrete	discrete-problems	4	Combinatorics	Counting & arrangements	Circular two-coloring with no isolated player
2024	Discrete	discrete-problems	5	Combinatorics	Probability	Loaded dice with equal sum distribution
2024	Discrete	discrete-problems	6	Number Theory	Digits & bases	Base 6 permutation numbers divisible by 43
2024	Discrete	discrete-problems	7	Combinatorics	Probability	Probability of GIGI substring in arrangement
2024	Discrete	discrete-problems	8	Number Theory	Arithmetic functions & valuations	Maximize ordered factorizations ratio over n
2024	Discrete	discrete-problems	9	Number Theory	Modular arithmetic	Sum of squares mod prime 83
2024	Discrete	discrete-problems	10	Combinatorics	Counting & arrangements	3x3 grid arrangements with adjacent sum repeats
2024	General	general-problems	1	Algebra	Arithmetic & word problems	Evaluate sum of products
2024	General	general-problems	2	Algebra	Arithmetic & word problems	Digit sum of two-digit odd addition

Year	Test	Source file	#	Subject	Topic	Key idea
2024	General	general-problems	3	Algebra	Arithmetic & word problems	Apples word problem, linear equations
2024	General	general-problems	4	Geometry	Triangles (lengths, angles, areas)	Perimeter of pie slice
2024	General	general-problems	5	Algebra	Arithmetic & word problems	Smallest fraction among list comparison
2024	General	general-problems	6	Number Theory	Digits & bases	Third-largest palindromic multiple of three
2024	General	general-problems	7	Combinatorics	Counting & arrangements	Arrangements of sandwich toppings with repeats
2024	General	general-problems	8	Geometry	Circles (tangents, power of a point, cyclic)	Circle through trisection points of square
2024	General	general-problems	9	Algebra	Equations & systems	Rational equations with equal values
2024	General	general-problems	10	Algebra	Sequences, series & recurrences	Arithmetic plus geometric sequences
2024	General	general-problems	11	Combinatorics	Counting & arrangements	Periodic coloring with usage limits
2024	General	general-problems	12	Algebra	Floor, ceiling & absolute value	Alternating fractions sum, greatest integer below
2024	General	general-problems	13	Number Theory	Divisibility, primes & factorization	Divisors of 44 with 24-n
2024	General	general-problems	14	Number Theory	Modular arithmetic	Powers of 3 plus n mod 5 counting
2024	General	general-problems	15	Algebra	Floor, ceiling & absolute value	Floor and ceiling equation, greatest x
2024	General	general-problems	16	Geometry	Polygons & area	Visibility region blocked by circle
2024	General	general-problems	17	Combinatorics	Counting & arrangements	Circular two-coloring with no isolated player
2024	General	general-problems	18	Geometry	Circles (tangents, power of a point, cyclic)	Internally tangent circles, ratio with point on circle
2024	General	general-problems	19	Number Theory	Digits & bases	Prefix divisibility with digits 0,2,4
2024	General	general-problems	20	Geometry	Circles (tangents, power of a point, cyclic)	Kite with circumcircle and incircle, area ratio
2024	General	general-problems	21	Combinatorics	Probability	Loaded dice with equal sum distribution
2024	General	general-problems	22	Algebra	Equations & systems	Symmetric three-variable quadratic system
2024	General	general-problems	23	Number Theory	Digits & bases	Base 6 permutation numbers divisible by 43
2024	General	general-problems	24	Algebra	Equations & systems	Triples with symmetric equation, AM-GM
2024	General	general-problems	25	Number Theory	Arithmetic functions & valuations	Maximize ordered factorizations ratio over n
2024	Geometry	geometry-problems	1	Geometry	Circles (tangents, power of a point, cyclic)	Three semicircles around equilateral triangle perimeter
2024	Geometry	geometry-problems	2	Geometry	Circles (tangents, power of a point, cyclic)	Circle through trisection points of square
2024	Geometry	geometry-problems	3	Geometry	Polygons & area	Visibility region blocked by circle, tangent lines
2024	Geometry	geometry-problems	4	Geometry	Circles (tangents, power of a point, cyclic)	Internally tangent circles, ratio with point on circle
2024	Geometry	geometry-problems	5	Geometry	Circles (tangents, power of a point, cyclic)	Kite with circumcircle and incircle, area ratio
2024	Geometry	geometry-problems	6	Geometry	Similarity, ratios & mass points	Orthic configuration areas, Ceva-like ratios
2024	Geometry	geometry-problems	7	Geometry	Coordinate & analytic geometry	Parallelogram altitudes, find CJ
2024	Geometry	geometry-problems	8	Geometry	3D geometry	Perpendicular chords in sphere, tetrahedron volume
2024	Geometry	geometry-problems	9	Geometry	Circles (tangents, power of a point, cyclic)	Incenter circle BIC, power of point from midpoint
2024	Geometry	geometry-problems	10	Geometry	Circles (tangents, power of a point, cyclic)	Incircle chord EF, poles, polar tools, area
2024	Guts	guts-problems	1	Geometry	Polygons & area	Nested squares, diagonal and distance DG
2024	Guts	guts-problems	2	Number Theory	Divisibility, primes & factorization	Smallest W reaching 2024 by doubling
2024	Guts	guts-problems	3	Number Theory	Divisibility, primes & factorization	Nested third-largest factors of integers
2024	Guts	guts-problems	4	Geometry	Circles (tangents, power of a point, cyclic)	Unicycle wheel circles, sector area
2024	Guts	guts-problems	5	Number Theory	Digits & bases	Smallest three-digit number not divisible by its digits

Year	Test	Source file	#	Subject	Topic	Key idea
2024	Guts	guts-problems	6	Algebra	Equations & systems	System of power equations, find (x+y)^100
2024	Guts	guts-problems	7	Geometry	Circles (tangents, power of a point, cyclic)	Shark fin area from sector and quarter circle
2024	Guts	guts-problems	8	Combinatorics	Games, invariants & pigeonhole	Cut tube so no triangle forms, Fibonacci-like
2024	Guts	guts-problems	9	Algebra	Logarithms & exponents	Iterated logarithm of tower function
2024	Guts	guts-problems	10	Number Theory	Diophantine equations	N=a^3+b^3-a^2b-ab^2 with six divisors
2024	Guts	guts-problems	11	Geometry	Triangles (lengths, angles, areas)	Right triangle with isosceles constructions, BD*BE
2024	Guts	guts-problems	12	Combinatorics	Inclusion-exclusion, bijections & lattice paths	Sum over upright triangles in Pascal triangle
2024	Guts	guts-problems	13	Algebra	Floor, ceiling & absolute value	Count solutions of x^floor(x)^{x}=n
2024	Guts	guts-problems	14	Number Theory	Digits & bases	Count n with digit sum minus product equals 5
2024	Guts	guts-problems	15	Combinatorics	Probability	Rotated parabola axis intersections probability
2024	Guts	guts-problems	16	Geometry	Triangles (lengths, angles, areas)	Orthocenter, midpoint, maximize area of DEF
2024	Guts	guts-problems	17	Number Theory	Arithmetic functions & valuations	Totient-like power ratio, weak integers sum
2024	Guts	guts-problems	18	Combinatorics	Probability	Die label flipping, probability sum hits k
2024	Guts	guts-problems	19	Algebra	Polynomials & Vieta	Quartic with pq=rs roots, evaluate
2024	Guts	guts-problems	20	Combinatorics	Probability	Random coin bias, binomial probability integral
2024	Guts	guts-problems	21	Geometry	Triangles (lengths, angles, areas)	Angle bisector, perpendicular bisector, area ratio
2024	Guts	guts-problems	22	Geometry	3D geometry	Tetrahedron with isosceles trapezoids, solid volume
2024	Guts	guts-problems	23	Algebra	Functions & functional equations	Count functions on subsets with modular property
2024	Guts	guts-problems	24	Algebra	Polynomials & Vieta	Minimal polynomial of 2cos(4pi/7) multiple, evaluate
2024	Guts	guts-problems	25	Other	Estimation	Count integers as sums of four Fibonacci numbers
2024	Guts	guts-problems	26	Other	Estimation	Choose c so iteration exceeds 2024 near step 20
2024	Guts	guts-problems	27	Other	Estimation	Conditional expected 6s, log of fractional part
2025	Algebra	algebra-problems	1	Algebra	Equations & systems	Linear combination of two equations
2025	Algebra	algebra-problems	2	Algebra	Arithmetic & word problems	Average with added 20 slips
2025	Algebra	algebra-problems	3	Algebra	Sequences, series & recurrences	Arithmetic sequence ratio and sum of terms
2025	Algebra	algebra-problems	4	Algebra	Functions & functional equations	Translate and rotate graph, roots of g
2025	Algebra	algebra-problems	5	Algebra	Equations & systems	Cube sum identities, symmetric system complex
2025	Algebra	algebra-problems	6	Algebra	Logarithms & exponents	Sine of log expression, sum of solutions
2025	Algebra	algebra-problems	7	Algebra	Complex numbers & roots of unity	Product of squared differences of fifth roots of unity
2025	Algebra	algebra-problems	8	Algebra	Logarithms & exponents	Change of base logs, simplify ratio
2025	Algebra	algebra-problems	9	Algebra	Complex numbers & roots of unity	Complex numbers with fifth power identity, abc values
2025	Algebra	algebra-problems	10	Algebra	Polynomials & Vieta	Power sum condition on ten complex numbers
2025	Calculus	calculus-problems	1	Calculus	Derivatives & optimization	Intersect f and f prime, solve quadratic
2025	Calculus	calculus-problems	2	Calculus	Derivatives & optimization	Maximize x^pi style function over positives
2025	Calculus	calculus-problems	3	Calculus	Limits	Limit of derivative at infinity
2025	Calculus	calculus-problems	4	Calculus	Limits	Limit of (2 arctan n/pi)^n
2025	Calculus	calculus-problems	5	Calculus	Advanced / tricky integrals	Integral with floor alternating signs
2025	Calculus	calculus-problems	6	Calculus	Limits	Expected binomial count limit, geometric rolls

Year	Test	Source file	#	Subject	Topic	Key idea
2025	Calculus	calculus-problems	7	Calculus	Limits	Limit of circle area fraction above parabola
2025	Calculus	calculus-problems	8	Calculus	Advanced / tricky integrals	Sign of integral with sin and gamma, count k
2025	Calculus	calculus-problems	9	Algebra	Inequalities & optimization	Riemann sum upper bound of cubic form
2025	Calculus	calculus-problems	10	Calculus	Advanced / tricky integrals	Integral of ln(sin^4+cos^4)
2025	Discrete	discrete-problems	1	Combinatorics	Probability	Two socks same color
2025	Discrete	discrete-problems	2	Number Theory	Divisibility, primes & factorization	n+10 multiple of n, factor of n+100
2025	Discrete	discrete-problems	3	Combinatorics	Counting & arrangements	Distribute identical blocks, positive, repeated count
2025	Discrete	discrete-problems	4	Combinatorics	Probability	Two-digit substrings divisible by three
2025	Discrete	discrete-problems	5	Number Theory	Arithmetic functions & valuations	gcd of lcms equals 2025, minimize sum
2025	Discrete	discrete-problems	6	Combinatorics	Counting & arrangements	Alternating multiples-of-three increasing sequences
2025	Discrete	discrete-problems	7	Combinatorics	Recursion & states	Seat permutations to adjacent seats on circle
2025	Discrete	discrete-problems	8	Number Theory	Arithmetic functions & valuations	Sum of divisors with even sigma, count h odd
2025	Discrete	discrete-problems	9	Combinatorics	Graphs, grids, colorings & tilings	Max edges with distinct degrees nonadjacent
2025	Discrete	discrete-problems	10	Number Theory	Diophantine equations	Perfect square 3^p*21^q+9*2^r, sum over triples
2025	General	general-problems	1	Algebra	Arithmetic & word problems	Evaluate perfect square expression
2025	General	general-problems	2	Geometry	Polygons & area	Square and triangle with equal perimeter
2025	General	general-problems	3	Number Theory	Digits & bases	Two-digit number with digit constraints
2025	General	general-problems	4	Combinatorics	Counting & arrangements	Outfits with forbidden pairs
2025	General	general-problems	5	Algebra	Arithmetic & word problems	Minimum moves multiply 3 or add 20
2025	General	general-problems	6	Algebra	Arithmetic & word problems	Relative speed on circular track
2025	General	general-problems	7	Geometry	Triangles (lengths, angles, areas)	Right triangle with square, perimeter of pentagon
2025	General	general-problems	8	Combinatorics	Probability	Two socks same color
2025	General	general-problems	9	Algebra	Arithmetic & word problems	Average with added 20 slips
2025	General	general-problems	10	Number Theory	Digits & bases	Digits product equals 2025
2025	General	general-problems	11	Algebra	Equations & systems	Radical equation solve
2025	General	general-problems	12	Number Theory	Divisibility, primes & factorization	n+10 multiple of n, factor of n+100
2025	General	general-problems	13	Algebra	Sequences, series & recurrences	Arithmetic sequence ratio and sum
2025	General	general-problems	14	Geometry	Polygons & area	Square divided into rectangles with areas, possible A
2025	General	general-problems	15	Combinatorics	Probability	Dice stickers probabilities
2025	General	general-problems	16	Geometry	Polygons & area	Square, pentagon, hexagon on common side, angle
2025	General	general-problems	17	Combinatorics	Counting & arrangements	Distribute identical blocks, positive, repeated count
2025	General	general-problems	18	Algebra	Arithmetic & word problems	Percent changes from digit replacement
2025	General	general-problems	19	Geometry	Polygons & area	Reflected rectangle forms rhombus, perimeter
2025	General	general-problems	20	Geometry	Triangles (lengths, angles, areas)	Equilateral triangle, perpendicular cevians, ratio
2025	General	general-problems	21	Combinatorics	Probability	Two-digit substrings divisible by three
2025	General	general-problems	22	Number Theory	Arithmetic functions & valuations	gcd of lcms equals 2025, minimize sum
2025	General	general-problems	23	Geometry	Polygons & area	Quadrilateral with distances to lines, area
2025	General	general-problems	24	Algebra	Logarithms & exponents	Sine of log expression, sum of solutions

Year	Test	Source file	#	Subject	Topic	Key idea
2025	General	general-problems	25	Combinatorics	Recursion & states	Count factor trees of 120
2025	Geometry	geometry-problems	1	Geometry	Triangles (lengths, angles, areas)	Right triangle with square, perimeter of pentagon
2025	Geometry	geometry-problems	2	Geometry	Polygons & area	Sheep tethered at corner, grazing area
2025	Geometry	geometry-problems	3	Geometry	Polygons & area	Square, pentagon, hexagon on common side, angle
2025	Geometry	geometry-problems	4	Geometry	Triangles (lengths, angles, areas)	Equilateral triangle, perpendicular cevians, ratio
2025	Geometry	geometry-problems	5	Geometry	Polygons & area	Quadrilateral with distances to lines, area
2025	Geometry	geometry-problems	6	Geometry	Transformations, loci & advanced tools	Locus of centers of tangent circles, ellipse area
2025	Geometry	geometry-problems	7	Geometry	3D geometry	Prism lattice points, Pick-type volume
2025	Geometry	geometry-problems	8	Geometry	Triangles (lengths, angles, areas)	Orthocenter maximize angle, find AB+AC
2025	Geometry	geometry-problems	9	Geometry	Circles (tangents, power of a point, cyclic)	Circumcircles of midpoint triangles, radical axis, AK squared
2025	Geometry	geometry-problems	10	Geometry	Circles (tangents, power of a point, cyclic)	Angle bisectors meeting circumcircles, XY parallel BC
2025	Guts	guts-problems	1	Geometry	Polygons & area	Perimeter of union of three squares
2025	Guts	guts-problems	2	Algebra	Arithmetic & word problems	Burger rate word problem system
2025	Guts	guts-problems	3	Algebra	Arithmetic & word problems	Arrange digits to minimize product expression
2025	Guts	guts-problems	4	Number Theory	Divisibility, primes & factorization	Six distinct factors of 30 with average 6
2025	Guts	guts-problems	5	Number Theory	Digits & bases	Digit sum operations then squared equals 2025
2025	Guts	guts-problems	6	Number Theory	Digits & bases	Multiples of 15 with digit sum above 15
2025	Guts	guts-problems	7	Combinatorics	Counting & arrangements	Coloring six squares with adjacency constraints
2025	Guts	guts-problems	8	Geometry	Triangles (lengths, angles, areas)	Right triangle with rectangle, area ratio, find CD
2025	Guts	guts-problems	9	Algebra	Sequences, series & recurrences	Arithmetic and geometric sequences, equal terms
2025	Guts	guts-problems	10	Combinatorics	Probability	Sum of two dice equal to winning numbers
2025	Guts	guts-problems	11	Algebra	Floor, ceiling & absolute value	Cubic with floor equation, sum of cubes of solutions
2025	Guts	guts-problems	12	Geometry	3D geometry	Pyramid with lateral edge angle, base area
2025	Guts	guts-problems	13	Algebra	Arithmetic & word problems	Nonassociative operation, max minus min over bracketings
2025	Guts	guts-problems	14	Algebra	Equations & systems	Exponent equals one, cases of base
2025	Guts	guts-problems	15	Geometry	Similarity, ratios & mass points	Cevian areas ratios, area of triangle PQR
2025	Guts	guts-problems	16	Number Theory	Modular arithmetic	Digitwise units product operation, count a
2025	Guts	guts-problems	17	Geometry	Polygons & area	Rectangle with maximized quadrilaterals, area
2025	Guts	guts-problems	18	Calculus	Limits	Palindromes in base 2025, smallest sqrt constant
2025	Guts	guts-problems	19	Combinatorics	Expected value	Expected area of quadrilateral from random points
2025	Guts	guts-problems	20	Geometry	Coordinate & analytic geometry	Lines and parabola, count regions
2025	Guts	guts-problems	21	Combinatorics	Graphs, grids, colorings & tilings	Connected graph by gcd, minimal trips
2025	Guts	guts-problems	22	Combinatorics	Expected value	Expected flips with reset Markov process
2025	Guts	guts-problems	23	Algebra	Sequences, series & recurrences	Product of 1+1/a_j equals 2025
2025	Guts	guts-problems	24	Geometry	3D geometry	Volume of solid from net
2025	Guts	guts-problems	25	Geometry	Circles (tangents, power of a point, cyclic)	Cyclic pentagon, inner pentagon area
2025	Guts	guts-problems	26	Other	Estimation	Count reducible short polynomials
2025	Guts	guts-problems	27	Other	Estimation	Sum of Euclidean algorithm step counts